A Public Normal-Form Framework for Entropy Bookkeeping in Sunflower Bounds

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Abstract

We describe a public normal-form framework for entropy-based approaches to the three-petal sunflower problem. The framework organizes tuple-level stopping rules, multi-sample projected shadows, conditioned copying with public transcript cost, deterministic DRC block selection, bounded active reservoirs, and delayed-window potential bookkeeping into a single finite-state interface. The main formal statement is an assembly theorem: assuming the local exits satisfy the publicness, copying, pruning, and ledger certificates specified below, the terminal shadow obeys the entropy criterion needed for a constant-base rank recurrence for balanced two-collision codes. Together with the standard random-coloring reduction, a complete verification of these interfaces would imply a constant-base bound for three-petal sunflower-free set families. The purpose of the manuscript is to isolate the proposed alphabet-free fixed-window mechanism and to make the remaining verification obligations explicit.

Contents

1 Introduction and Conditional Reduction

This paper develops a public-normal-form route to a possible constant-base sunflower bound. The emphasis is on the method and on the interfaces that a complete proof would have to verify, not on claiming an unconditional settlement of the sunflower conjecture. The central assembly statement is an interface theorem: once all local exits are available in the stated public normal form, the global entropy bookkeeping closes.

Sunflowers, or Δ-systems, were introduced by Erdos and Rado [10]. Their classical sunflower lemma gives a bound of order $k!(r-1)^k$ for r-petal sunflowers in k-uniform families, and they conjectured that for each fixed r this can be replaced by C_r^k. For three petals, the conjecture remains the relevant benchmark for constant-base entropy or encoding arguments. Kostochka obtained the first asymptotic improvement over the Erdos–Rado bound [14]. The modern breakthrough was the spread/robust-sunflower method of Alweiss, Lovett, Wu, and Zhang [3]; subsequent entropy and probabilistic presentations by Rao [19], Tao [23], and Bell, Chueluecha, and Warnke [5] gave the current logarithmic-base form of the method. Related threshold work of Frankston, Kahn, Narayanan, and Park [12] shows how robust sunflower ideas interact with expectation-threshold problems. Classical lower-bound constructions and finite-universe variants also play an important role in the surrounding picture [1, 11].

There is also a bounded-alphabet route through the polynomial method and slice rank. The Croot–Lev–Pach and Ellenberg–Gijswijt cap-set breakthroughs [7, 9] and Tao's slice-rank formulation [22], together with the sunflower-free applications of Alon, Shpilka, and Umans [4] and Naslund–Sawin [17], show that strong bounds are available when the alphabet is controlled. Those arguments, however, retain an alphabet-dependent base. The present framework is meant to separate a different question: whether a public transcript system can route all large-alphabet information either to terminal projected shadows or to bounded local ledgers, leaving the active recurrence alphabet-free.

The technical language used below is standard. Entropy, relative entropy, data processing, Pinsker-type estimates, and chain rules are used in the usual information-theoretic sense [6, 8, 15, 18, 21]; Renyi entropy enters only through conditioned-copying estimates [20]. The random-coloring reduction is a first-moment probabilistic-method step [2], and the DRC blocks are finite coordinate versions of dependent random choice [13].

Self-contained interface convention

For the framework to be externally checkable, every statement used in the conditional assembly must be one of the following:

(1) stated and proved in this manuscript;

(2) cited as a standard external theorem in the ordinary mathematical way.

The elementary, structural, product, residual, old-support, and bookkeeping interfaces used below are therefore stated explicitly. The reduction from the original set-family formulation to balanced two-collision codes is also included, so that the conditional conclusion can be read in the original sunflower language once the interface hypotheses have been verified.

Proposed fixed-window mechanism

The candidate alphabet-free mechanism is the fixed, labelled hash window of Theorem 9.2. In a usual random-restriction or encoding argument, a useful coordinate is often found together with a value or a long restriction transcript; exposing that value costs on the order of the available alphabet entropy. The proposed normal form separates the two operations. A high-mass value is terminalized immediately as a projected shadow and is paid by pinned coordinates. A low-collision coordinate is not exposed by value: it is placed into a fixed hash window for one public sample label. After only h surviving coordinates, with h fixed once and for all, the window returns a distinct near-sunflower block at cost

$$O(B_{\mathrm{win}} + h),$$

including the public transcript, the Renyi conditioned-copy tax, and the exact split exposure. This cost depends on the fixed thresholds, not on the alphabet sizes or on k. Repetition is controlled separately by the fresh/old/scale potentials, so the hash window is not an additional reusable budget. This is the proposed point at which alphabet-dependent value exposure would be removed from the active recurrence, if the local hash and ledger interfaces are fully verified.

Key mechanism trace

The framework uses the fixed window only as one local progress source, not as a replacement for all near-sunflower production. The branch accounting is as follows.

First, a positive-KL or collision output never exposes an alphabet value unless it terminalizes as a projected shadow. Otherwise it contributes one low-collision coordinate to the labelled hash queue. The hash queue has a fixed cumulative transcript budget B_{win}: prefix, tail/pruning, and coordinate-production transcript are appended only while the cumulative cost remains below that budget. If the next set-valued tail/pruning increment would exceed the budget, it is not appended; the branch terminalizes on the already paid value prefix. Prefixes which exceed the remaining budget are terminal projected shadows and do not enter the hash queue.

Second, the normal-form hash window is applied only after enough distinct coordinates for one public sample label have accumulated before the cumulative budget closes. If the budget closes first, the branch is a terminal or bounded-support exit and is charged by the same fixed hash ledger. In the actual transition system the destruction budget in Theorem 9.2 is zero: whenever a produced coordinate later has an atom of mass at least β, the branch stops as a value shadow. Thus a nonterminal window with at least $2h$ produced coordinates has h surviving coordinates deterministically; the Markov form of Theorem 9.2 is kept only to make the fixed-window statement robust.

Third, on those h surviving coordinates the one-sample conditional law has coordinate collision at most β and codeword atoms below β. Hence three independent conditioned copies are distinct and have at most $6\beta h$ split coordinates with fixed positive probability. The Renyi copying tax and the exact split exposure cost

$$O(B_{\mathrm{win}} + h),$$

with no dependence on the alphabet sizes or on $|\mathcal{C}|$.

Finally, the returned split block is processed by the residual decision order. Fresh coordinates can be promoted only once. Old incidences are recorded by raw revisit counters and charged to the weighted debit N_{old}; the raw counters are capped by the high-revisit rule. Scale-only returns decrease active scale geometrically, and every new scale is born with a paid certificate. Consequently repeated use of fixed windows is paid by the fresh, old, and scale potentials; there is no additional union bound over the whole code family.

Theorem 1.1 (Public normal-form interface reduction)**.** *Assume that the public-normal-form transition system satisfies the local interfaces stated in Sections 6–18: public tuple exits, fixed-window hash returns, residual repair, old-window projection, bounded reservoirs, pruning normalization, global charge summation, and scale-birth certificates. Then there is a constant $C < \infty$, depending only on the fixed hierarchy of parameters, such that every balanced two-collision code of length k started from the root state of Lemma 3.3 has size at most C^k. Consequently, a complete verification of these interfaces in the intended sunflower setting would give that every rank at most k set family with no three-petal sunflower has size at most $(eC)^k$, after increasing the constant for the finitely many small values of k.*

Proof. Let $\mathcal{C}$ be a balanced two-collision code of length k and let μ be the uniform law on $\mathcal{C}$. By Lemma 3.3, this gives an admissible public root state with rank k, empty old support, and root potential bounded by $O(k)$.

Run the normal-form transition system of Definition 3.5 with the constants chosen as in Section 18. Under the stated interface hypotheses, every nonterminal mode has an admissible public exit: close-pair exits are handled by Theorem 12.4, product-KL exits by Proposition 12.6, positive-KL outputs by Theorem 12.5, public hash returns by Theorem 9.2 and Propositions 9.5 and 9.6, residual repair by Proposition 11.2 and Lemma 10.3, old-supported repair by Lemma 13.5 and Proposition 10.8, old windows by Theorem 14.3, bounded reservoirs by Proposition 15.3, and active scale birth by Proposition 17.2. Therefore Theorem 16.6 gives a terminal projected shadow whose entropy is controlled by pinned coordinates, sample labels, pruning loss, and a telescoping potential.

Theorem 5.2 converts this entropy bound into a shadow atom with the required mass. The strengthened induction absorbs the potential drop, and Theorem 4.2 gives

$$|\mathcal{C}| \le D^k \exp(A\Phi(\mathsf{S}_0)).$$

Since the root has empty old support and

$$\Phi(\mathsf{S}_0) \le (B_{\mathrm{rk}} + B_f + B_s)k,$$

the right-hand side is at most

$$\big(D\exp(A(B_{\mathrm{rk}} + B_f + B_s))\big)^k.$$

Taking this quantity as C^k proves the balanced-code bound. Finally Theorem 2.4 transfers the bound back to ordinary rank at most k set families, with the factor e^k coming from random coloring. $\square$

Remark 1.2. *Theorem 1.1 is a conditional bookkeeping theorem. Its role is to separate the global entropy accounting from the local mathematical interfaces. The rest of the manuscript records a proposed verification of those interfaces in public normal form, replacing private terminal transcripts by projected shadows, potential drops, pruning charges, or fixed public hash windows.*

2 Reduction to Balanced Two-Collision Codes

A three-petal sunflower is a triple of distinct sets A, B, C satisfying

$$A \cap B = A \cap C = B \cap C.$$

The rank of a family is the maximum size of one of its members.

Definition 2.1 (Balanced two-collision code)**.** *A code $\mathcal{C} \subseteq \prod_{i=1}^k \Omega_i$ is a balanced two-collision code of length k if for every three distinct codewords $x, y, z \in \mathcal{C}$ there is a coordinate i such that exactly two of x_i, y_i, z_i are equal.*

Lemma 2.2 (Padding to uniformity)**.** *Let $\mathcal{F}$ be a rank at most k family with no three-petal sunflower. Then there is a k-uniform family $\mathcal{F}^+$ with $|\mathcal{F}^+| = |\mathcal{F}|$ and no three-petal sunflower.*

Proof. For each $A \in \mathcal{F}$, add a private set D_A of $k - |A|$ new vertices, disjoint from all old vertices and from all D_B with $B \ne A$, and set $A^+ = A \cup D_A$. The family $\mathcal{F}^+ = \{A^+ : A \in \mathcal{F}\}$ is k-uniform and has the same cardinality as $\mathcal{F}$.

If A^+, B^+, C^+ formed a sunflower, then every element in any pairwise intersection of two of them would be an old vertex, because each dummy vertex belongs to exactly one padded set. Hence

$$A^+ \cap B^+ = A \cap B, \quad A^+ \cap C^+ = A \cap C, \quad B^+ \cap C^+ = B \cap C.$$

Thus A, B, C would form a sunflower in $\mathcal{F}$, a contradiction. $\square$

Lemma 2.3 (Random coloring to a partite code)**.** *Let $\mathcal{F}$ be a k-uniform family with no three-petal sunflower. Then there is a balanced two-collision code $\mathcal{C}$ of length k with*

$$|\mathcal{C}| \ge \frac{k!}{k^k}\,|\mathcal{F}|.$$

Proof. This is the standard random-coloring first-moment argument; see [2]. Color the ground set of $\mathcal{F}$ independently and uniformly by colors $1, \ldots, k$. A fixed k-set is colorful, meaning that it receives each color exactly once, with probability $k!/k^k$. Therefore some coloring has a colorful subfamily $\mathcal{F}_0 \subseteq \mathcal{F}$ with $|\mathcal{F}_0| \geq (k!/k^k)|\mathcal{F}|$.

For this coloring, let Ω_i be the set of vertices of color i which appear in $\mathcal{F}_0$. Encode every $A \in \mathcal{F}_0$ by the word $x(A) = (x_1(A), \ldots, x_k(A))$, where $x_i(A)$ is the unique vertex of color i in A. This gives an injective map from $\mathcal{F}_0$ into $\prod_i \Omega_i$; let $\mathcal{C}$ be its image.

Take three distinct words $x, y, z \in \mathcal{C}$ and their corresponding colorful sets $A, B, C \in \mathcal{F}_0$. In a fixed color class i, the three symbols x_i, y_i, z_i are either all equal, all distinct, or exactly two are equal. If every coordinate is all equal or all distinct, then the contribution of each color class to the three pairwise intersections is the same; hence A, B, C form a sunflower. Since $\mathcal{F}_0$ is sunflower-free, some coordinate has exactly two equal symbols. Thus $\mathcal{C}$ is a balanced two-collision code. □

Theorem 2.4 (Balanced-code reduction)**.** *Suppose that every balanced two-collision code of length k has size at most C^k for an absolute constant C. Then every rank at most k family with no three-petal sunflower has size at most $(eC)^k$.*

Proof. By Lemma 2.2, it is enough to consider a k-uniform sunflower-free family $\mathcal{F}$. Lemma 2.3 gives a balanced two-collision code $\mathcal{C}$ of length k with

$$|\mathcal{C}| \geq \frac{k!}{k^k}|\mathcal{F}|.$$

The assumed code bound gives

$$|\mathcal{F}| \leq \frac{k^k}{k!}C^k.$$

Using $k! \geq (k/e)^k$, we obtain $|\mathcal{F}| \leq (eC)^k$. □

3 Encoded Codes and Balanced States

We work with a finite ambient code

$$\mathcal{C} \subseteq \prod_{i=1}^{k} \Omega_i$$

and with public active coordinate blocks $I \subseteq [k]$. A random codeword under a law μ is denoted by X. At a recursive cylinder state the pinned coordinates have been removed from the ambient product, and the remaining coordinates are relabelled as the current ambient set. Thus every later occurrence of $[k]$ inside a state, including $[k] \setminus U$ in the potential, means the current unpinned ambient coordinate set, not necessarily the original root set.

Definition 3.1 (Admissible two-collision state)**.** *A state is a tuple*

$$\mathsf{S} = (I, \mathcal{C}, \mu, U, \mathcal{P}, \mathcal{R}),$$

where I is the active coordinate set, $U \subseteq [k]$ is the global old support, $\mathcal{P}$ is the public state, and $\mathcal{R}$ is a bounded active reservoir of sampled codewords. The code $\mathcal{C}$ is a balanced two-collision code after the public value pins already present in $\mathcal{P}$. Heavy coordinate fibres are immediate exits of the proof tree, not additional entry assumptions.

The final proof must be insensitive to analysis-only data. Thus we separate public state from private transcript. Formally, each node v carries a public sigma-algebra $\mathcal{F}_v$ generated by the fixed constants, the root coordinate order, and all previously paid transcript variables. A block, law, tie-break, or transition selector is public at v if it is $\mathcal{F}_v$-measurable. If a non-public child selector Γ_v is revealed, then the tree ledger pays $\mathbf{H}(\Gamma_v \mid E_v)$ and the child public sigma-algebra is

$$\mathcal{F}_{v'} = \mathcal{F}_v \vee \sigma(\Gamma_v)$$

on that child event. Variables used only to analyze the branch are not adjoined to the terminal sigma-algebra unless they are paid in this way or are terminalized as part of a projected shadow.

Definition 3.2 (Public normal form)**.** *A proof node is in public normal form if:*

(i) *every coordinate block used for a hash, DRC, close-pair, or near-sunflower transition is $\mathcal{F}_v$-measurable after the already paid public transcript at that node;*

(ii) *every value pin terminalizes immediately as a projected value-shadow; if the pin is selected by a high-mass existence rule rather than by exposing a full value selector, the retained atom is chosen by the fixed public tie-break among atoms above the threshold and the discarded complement is charged as a pruning-normalization loss; no part of that complement is inherited by a nonterminal child;*

(iii) *every DRC block is selected by a deterministic tie-break from the current public law;*

(iv) *every hash window carries one public active sample label, and all hash coordinates accumulated in that window are ambient coordinates of that same sample label under the paid tuple event; the fixed budget B_{win} is a per-labelled-window budget, not a global per-branch budget, and a different sample label opens a different window with its own stopped transcript;*

(v) *every old-window queue carries one public sample-label signature, namely the ordered active triple/reservoir labels whose equality pattern is being recorded; old-pattern atoms are nested only while this signature is unchanged;*

(vi) *nonterminal nodes carry only a bounded active reservoir;*

(vii) *every conditional one-sample law used at a tuple node is the law of an active sample label under a paid tuple event, and every conditioned copy is realized back in independent root samples by Lemma 8.4 or Lemma 8.5 before it can contribute to a terminal shadow;*

(viii) *private analysis data either never enters the terminal sigma-algebra, is charged to a potential drop or pruning event, or is terminalized into a projected shadow.*

Lemma 3.3 (Root state)**.** *Let $\mathcal{C} \subseteq \prod_{i=1}^{k} \Omega_i$ be a balanced two-collision code and let μ be the uniform law on $\mathcal{C}$. Then*

$$\mathsf{S}_0 = ([k], \mathcal{C}, \mu, \varnothing, \mathcal{P}_0, \varnothing)$$

is an admissible public root state, where $\mathcal{P}_0$ contains only the fixed coordinate order and the fixed threshold constants.

Proof. There are no old coordinates, no private transcript, and no active reservoir at the root. The coordinate order and constants are deterministic, hence public. The defining two-collision property of $\mathcal{C}$ is exactly the admissibility condition before any public value pins are made. $\square$

Lemma 3.4 (Root activation dichotomy)**.** *Let μ be a public law on a balanced two-collision code on an active block I, $|I| = m$. Fix constants $0 < \alpha < 1$ and $0 < \tau < 1$. For independent $X, Y, Z \sim \mu$, at least one of the following occurs:*

(i) *terminal atom: either a codeword has μ-mass at least $1/12$, or some coordinate value has $\mu(X_i = a) \geq \tau$;*

(ii) *near-sunflower activation:*

$$\mathbb{P}[X, Y, Z \text{ distinct and } |\operatorname{Split}_I(X, Y, Z)| \leq 6\alpha m] \geq 1/4;$$

(iii) *close-pair activation: for every integer*

$$1 \leq s \leq (1-\tau)\alpha m/4$$

there is a public block $J \subseteq I$, $|J| = s$, such that

$$\mathbb{P}[X_J = Y_J,\ Z_j \neq X_j\ (j \in J)] \geq \frac{(1-\tau)\alpha}{2}\left(\frac{(1-\tau)\alpha}{4}\right)^s.$$

Proof. If the terminal atom alternative holds, choose the first such atom in the fixed public order, retain that atom as a projected terminal shadow, and charge the constant pruning-normalization loss of the retained atom as in the public normal-form convention. No complement branch is inherited by a nonterminal child. Assume from now on that no terminal atom alternative holds. In particular,

$$\max_x \mu(x) < 1/12, \qquad \max_a \mathbb{P}[X_i = a] < \tau \quad (i \in I).$$

If the average collision on I is at most α, then Lemma 6.2 gives

$$\mathbb{P}[|\operatorname{Split}_I(X, Y, Z)| \leq 6\alpha m] \geq 1/2.$$

The probability that X, Y, Z are not all distinct is at most $3\sum_x \mu(x)^2 \leq 3\max_x \mu(x) < 1/4$. Hence the near-sunflower activation has mass at least $1/4$.

It remains to consider the case

$$\frac{1}{m}\sum_{i \in I} \operatorname{Coll}_i(\mu) > \alpha.$$

For a coordinate i, write $p_a = \mathbb{P}[X_i = a]$. Then

$$\mathbb{P}[X_i = Y_i \neq Z_i] = \sum_a p_a^2(1-p_a) \geq (1-\tau)\sum_a p_a^2 = (1-\tau)\operatorname{Coll}_i(\mu).$$

Thus the events $E_i = \{X_i = Y_i \neq Z_i\}$ have average mass at least $(1-\tau)\alpha$. Apply Lemma 10.1 to the probability space of triples and the events E_i. The block is chosen by the fixed public tie-break among all valid DRC blocks, so it is public. □

Definition 3.5 (Normal-form transition system)**.** *The proof tree is the smallest adaptive tree generated by the following public modes. A fixed integer m_0 is chosen in the constant hierarchy; any node whose active rank and all active block sizes are $< m_0$ is declared a bounded leaf.*

(1) *Inactive code-law mode. The node has an admissible public state but no active block. Apply Lemma 3.4. Terminal atoms stop, near-sunflower activations enter residual mode, and close-pair activations enter close-pair mode after choosing, for active rank $m \geq m_0$, the first valid DRC block of size*

$$\lfloor (1-\tau)\alpha m/8 \rfloor.$$

(2) *Close-pair mode. The node carries a public block J and a tuple law with $X_J = Y_J = A$ and $Z_j \neq A_j$ on J. Apply Theorem 12.4. Low-entropy and atom exits stop as projected shadows; KL/collision exits route through terminal projected tuple pins, charged pruning, or hash mode; the law-level near-sunflower exit enters residual mode.*

(3) *Hash-window mode. The node carries a public sample label, a public hash transcript prefix whose length is bounded by the per-window budget B_{win}, and a bounded list of public low-collision coordinates for that label. Each active sample label has its own window; a coordinate produced for another label is routed to that label's separate stopped window, subject to the same per-window budget and the single-token rules below. Apply Propositions 9.5 and 9.6, and Theorem 9.2. The output is a terminal value shadow, a bounded-support exit, or a public near-sunflower block which enters residual mode.*

(4) *Residual mode. The node carries a public near-sunflower block and an exact split exposure. Apply the public order of Proposition 10.5. The output is fresh progress, scale descent, a close-pair block, a bounded leaf, or an old-window block.*

(5) *Old-window mode. The node carries public old blocks of one scale and one public sample-label signature. Apply Theorem 14.3. The output is a terminal low-symbol shadow, a high-revisit terminal/pruning/hash exit, or a low-revisit continuation whose non-value transcript has been registered in the old-incidence ledger. Bounded short windows remain in old-window mode through the delayed queue.*

(6) *Terminal mode. A terminal node outputs a projected shadow and projects away all unpaid analysis-only data before the final entropy count.*

(7) *Bounded-leaf mode. A bounded leaf is counted by the crude constant D_0^{rank} after D_0 is chosen large enough for the finitely many ranks and block sizes below m_0. The required finite bounds come from the classical Erdos–Rado sunflower bound in these finitely many fixed ranks.*

All coordinate and block choices in these rules are deterministic functions of the public state and the fixed thresholds, except for value atoms that are immediately terminalized.

Proposition 3.6 (Transition system is public and exhaustive). *Every nonterminal node generated by Definition 3.5 is in public normal form and is in exactly one of the five nonterminal modes listed there. The entry conditions for these modes are disjoint: inactive nodes have no active block, close-pair nodes carry a certified pair block, hash-window nodes process one selected sample-label window and no pending returned hash token, residual nodes carry an exact split block returned by activation, hash, or close-pair, and old-window nodes carry old blocks of one scale and one sample-label signature. If several labelled hash windows are available, the next active window is chosen by the fixed public order; each labelled window has its own stopped transcript and its own possible token. The listed exit alternatives for the active mode form an exhaustive partition after deterministic public tie-breaks. Every child is either terminal or again belongs to one of those modes.*

Proof. Induct on tree depth. The root is public by Lemma 3.3. At each step the cited local result is used as an exhaustive list of children, not as a diagnostic after the child has been chosen.

In inactive mode, Lemma 3.4 uses only the current public law and fixed tie-breaks. Its alternatives are terminal atom, residual near-sunflower, and close-pair activation; if the terminal atom condition holds it is taken first, otherwise the collision average dichotomy separates the near-sunflower and close-pair cases. The chosen close-pair block has the fixed-fraction size displayed in

Definition 3.5; the choice of m_0 makes this size nonzero and within the DRC range of Lemma 3.4. In close-pair mode, the input block is public by the mode definition and Theorem 12.4 returns only terminal low-entropy/atom exits, product-KL exits routed by Proposition 12.6, collision exits routed exhaustively by Lemma 6.9, or a law-level near-sunflower residual return; the theorem's decision order makes these exits a partition. In hash-window mode, the active state is one selected labelled window. Coordinates for another sample label are placed in that label's separate stopped window and are not merged into the current transcript. Theorem 9.2 requires coordinate production to be public for the fixed label of the active window and pays that window's public transcript before any copying, while Propositions 9.5 and 9.6 account for terminal pins, bounded support, survival scans, failed accumulation, and residual returns.

In residual mode, the exact split block is already public. The residual decision order first tests fresh support, then old support, then sparse scale/close-pair alternatives; DRC blocks are public by Definition 11.1, fresh support promotion is public by Lemma 10.3, and the remaining old-supported witnesses are routed by Lemma 13.5 and Proposition 10.8 to scale, old-window, or close-pair modes. In old-window mode, Theorem 14.3 either terminalizes a low-symbol old value, exits by high revisit, or registers all non-value old data before continuation; Proposition 14.4 delays only bounded short windows of the same public scale and signature. Terminal nodes keep only projected shadows, and bounded leaves are counted by the finite constant D_0.

The tuple-node realization invariant is preserved because value pins stop immediately, while the only operation which creates new conditioned samples is conditioned copying, and Lemmas 8.4 and 8.5 realize those copies in independent root samples with the displayed transcript cost. Thus public normal form and the mode classification are preserved. □

Proposition 3.7 (Branch invariants). *On every retained branch of the normal-form transition system the following invariants hold.*

(i) *The public sigma-algebras are increasing, and every increase is generated by a child selector whose conditional entropy is charged in the proof-tree ledger.*

(ii) *Each labelled hash window has its own stopped transcript budget B_{win} and one public sample label. Different sample labels have different windows. The global branch is not charged by a shared hash transcript budget; it is charged by the number of completed or bounded labelled windows, each of which is later terminalized or assigned to one hard exit.*

(iii) *Every tuple law is the law of a fixed finite list of independent root samples conditioned on the paid transcript accumulated on the branch. New conditioned samples are introduced only through Lemmas 8.4 and 8.5.*

(iv) *The old support U is monotone increasing. Fresh promotions are disjoint subsets of $[k] \setminus U$, and each promoted coordinate is removed from the fresh potential exactly once.*

(v) *Old-window incidences increase the raw revisit counters monotonically until the high-revisit exit is triggered, and each registered incidence debits the weighted old account N_{old}. No low-revisit continuation uses an incidence which has not been registered and debited.*

(vi) *A hash return carries at most one pending token, of value at most C_{hash}, attached to the returned block. The transition system processes that returned block until the token is assigned to the first subsequent terminal, fresh, old, scale, or birth exit; before this assignment no child state, descendant block, or new hash window is entered from that block. If the first exit is deletion-trapped scale descent or a birth transition, the token is assigned at the parent residual transition and is not inherited by the child.*

(vii) *Every active block is born with the certificate of Definition 17.1. After birth, deletion-trapped returns decrease its scale by the fixed factor in Lemma 10.9; no transition increases active scale without a new birth certificate.*

(viii) *Old-pattern atoms are nested under growth of U for a fixed sample-label signature. When U is enlarged to U^+ without changing that signature, any later pattern atom on U^+ is refined inside the previously paid atom on U and exposes only the pattern on $U^+ \setminus U$ as new data. If the active triple/reservoir signature changes, every bounded old queue for the old signature is first registered, terminalized, or charged to the hard exit which changes the signature, and then cleared.*

Proof. The first two invariants are exactly the public-normal-form definition and the copying clauses in Proposition 3.6. Fresh promotion is performed only by Lemma 10.3 and by bounded hash-support exits, both of which add public subsets of the current complement $[k] \setminus U$ to U; hence the promoted sets are disjoint along a branch. Old incidences are registered in Lemma 13.5, Theorem 14.3, and Proposition 14.4; these are precisely the old-window continuations. Each registration increments the raw revisit counter and debits the weighted old account, while the high-revisit exit stops the low-revisit regime when the raw counter reaches the cap. Proposition 9.5 and Lemma 9.8 give the single-token and token-free-child rules for hash returns. Finally, Proposition 17.2 enumerates the sources of active blocks and assigns their certificates, while Lemma 10.9 gives the only scale-decreasing nonterminal continuation. These are the only nonterminal modes by Proposition 3.6. The nested old-pattern rule is a normal-form convention for refining public sigma-algebras for the same active sample-label signature: a pattern already paid on U remains part of the public state for that signature, and a later pattern on U^+ is the intersection of that old atom with a newly paid pattern on $U^+ \setminus U$. Signature-changing transitions cannot inherit that atom; Proposition 14.4 flushes the bounded unpaid queue before such a change is entered. $\square$

Proposition 3.8 (Finite truncation)**.** *After bounded leaves are imposed at scale m_0, every branch of the normal-form transition system has a finite truncation whose discarded tail has total mass bounded by the pruning estimates already charged in the transition system.*

Proof. The only zero-entropy transitions are deterministic public choices: fixed tie-breaks for blocks, mode changes after a charged selector has already been revealed, and bookkeeping relabellings such as unchanged-signature old-queue growth. Such a choice cannot repeat in the same state. It either enters a terminal/bounded leaf, enters a hash window with a fixed remaining transcript budget, enters an old queue whose cumulative incidence is counted, or enters an active block with a rank/scale certificate. Hence every infinite retained path would have to contain infinitely many charged or monotone-budget steps of one of the types listed below.

Fresh promotions and fresh/rank descents decrease $\Phi_{\mathrm{fresh}} + \Phi_{\mathrm{old}}$ by a fixed positive amount per promoted coordinate. Old-window incidences are either registered and debited in N_{old}, which decreases Φ_{old} by the weighted debit, or they trigger the high-revisit exit of Theorem 14.3; the delayed queue prevents bounded short windows from restarting without increasing the same cumulative counters. Hash windows have a fixed stopped transcript budget B_{win} and fixed target size $2h$: before reaching that target they either close as terminal or bounded-support exits, and after reaching it they return to residual mode with a token which must be assigned before any descendant window is opened. Thus hash mode cannot contribute infinitely many zero-cost continuations. Scale descents are geometric by Lemma 10.9; once the scale falls below m_0 the branch is a bounded leaf. Every new scale is born with a certificate by Proposition 17.2, so the total geometric scale available along a branch is bounded by the initial active-rank or parent active-scale account, fresh-promotion drops, and weighted old-debit budgets. Bounded hash tokens pay their fixed transcript charge only after assignment to a hard exit; they do not create recurrence potential. Terminal pins and public DRC certificates do not create recurrence potential.

The only discarded branches in the finite truncation are explicit pruning branches, such as rare positive-tail pruning, first-crossing light-complement pruning from Lemma 6.7, high-mass terminal atom pruning from the public normal-form convention, hash-survival pruning from Definition 9.1, and the public fresh-reset pruning of Lemma 14.2. Their total discarded mass is bounded by the estimates charged at the pruning step. In particular, positive-tail and light-complement pruning are charged inside the same prefix/tail transcript which then immediately produces a terminal pin or a hash coordinate. If the fixed tail/pruning increment would overflow the window, it is not retained as a set-valued terminal shadow; the branch stops on the value prefix already paid by the KL chain rule. Thus repeated prunings consume the fixed hash-window budget or terminate as projected value-shadows. Therefore truncating after all potential-drop, incidence, scale, and bounded-window budgets up to a fixed level discards only already charged pruned mass. □

Proposition 3.9 (Proof-tree entropy ledger)**.** *Let $\mathcal{T}$ be a finite adaptive proof tree. Let Γ be its full terminal leaf transcript, and let $\Theta = f(\Gamma)$ be the projected terminal shadow obtained after all unpaid analysis-only data are projected away. For an internal node v, let E_v be the event that the branch reaches v, and let A_v be the child selector at v. Then*

$$\mathbf{H}(\Theta) \le \mathbf{H}(\Gamma) = \sum_{v \in \mathcal{T}^\circ} \mathbb{P}(E_v)\, \mathbf{H}(A_v \mid E_v).$$

Consequently, if nonnegative local charges c_v satisfy

$$\mathbf{H}(A_v \mid E_v) \le \mathbb{E}[c_v \mid E_v] \qquad (v \in \mathcal{T}^\circ),$$

then

$$\mathbf{H}(\Theta) \le \mathbb{E} \sum_{v \text{ reached}} c_v.$$

Proof. Since Θ is a function of Γ, the data-processing inequality gives $\mathbf{H}(\Theta) \le \mathbf{H}(\Gamma)$. Reveal the terminal leaf by scanning the finite tree in a fixed topological order, so every parent is scanned before its children, and when a reached internal node is met, reveal its child selector. Selectors at unreached nodes are deterministic cemetery symbols. The chain rule gives

$$\mathbf{H}(\Gamma) = \sum_{v \in \mathcal{T}^\circ} \mathbf{H}(A_v \mathbf{1}_{E_v} \mid \text{previous selectors}).$$

Conditioned on the previous selectors, the event E_v is already known. Thus the contribution is zero off E_v and is $\mathbf{H}(A_v \mid E_v)$ on E_v, giving the displayed tree identity. Multiplying the local charge inequality by $\mathbb{P}(E_v)$ and summing over v gives the last claim. □

Proposition 3.10 (Pruning normalization charge)**.** *Suppose a finite proof tree is run with charged pruning steps. At a pruning step the current conditional law is restricted to a retained event of conditional mass $q_v > 0$, the discarded complement is not inherited by any recursive child, and the branch records the loss*

$$\ell_v = -\log q_v.$$

Let $\widehat{\mathbb{P}}$ be the normalized law on retained terminal outcomes, let

$$L_{\mathrm{pr}} = \sum_{v \text{ pruned on the branch}} \ell_v,$$

and let $\mathbb{P}_{\mathrm{orig}}$ denote the original, unnormalized mass of the retained terminal shadow event before pruning projection. Then

$$\widehat{\mathbb{E}}\big[-\log \mathbb{P}_{\mathrm{orig}}(\Theta)\big] \le \mathbf{H}_{\widehat{\mathbb{P}}}(\Theta) + \widehat{\mathbb{E}} L_{\mathrm{pr}}.$$

Consequently, if

$$\mathbf{H}_{\widehat{\mathbb{P}}}(\Theta) + \widehat{\mathbb{E}} L_{\mathrm{pr}} \le \widehat{\mathbb{E}} B$$

for a nonnegative branch charge B, then some retained terminal outcome ω with shadow $\theta = \Theta(\omega)$ satisfies

$$-\log \mathbb{P}_{\mathrm{orig}}(\Theta = \theta) \le B(\omega).$$

Proof. For a shadow atom θ, the original retained mass is

$$\mathbb{P}_{\mathrm{orig}}(\Theta = \theta) = \widehat{\mathbb{P}}(\Theta = \theta)\, \widehat{\mathbb{E}}\big[e^{-L_{\mathrm{pr}}} \mid \Theta = \theta\big]\,.$$

Therefore Jensen's inequality gives

$$-\log \mathbb{P}_{\mathrm{orig}}(\Theta = \theta) \le -\log \widehat{\mathbb{P}}(\Theta = \theta) + \widehat{\mathbb{E}}[L_{\mathrm{pr}} \mid \Theta = \theta]\,.$$

Averaging over θ proves the displayed expectation bound. Applying the same averaging argument to $-\log \mathbb{P}_{\mathrm{orig}}(\Theta) - B$ gives the final atom. □

4 Multi-Sample Projected Shadows

Definition 4.1 (Multi-sample value-shadow)**.** *At the recurrence interface, let $X^1, \ldots, X^r$ be independent samples from the uniform law on the code currently being bounded. Internal conditional tuple laws used by the proof tree are auxiliary devices; every terminal shadow is evaluated after pulling its event back to these root samples. A multi-sample projected value-shadow is a finite list*

$$\theta = ((J_s, a_s))_{s \in S},$$

where $\varnothing \ne S \subseteq [r]$, $\varnothing \ne J_s \subseteq I$, and $a_s \in \prod_{j \in J_s} \Omega_j$. A terminal event E is compatible with θ if

$$E \subseteq \bigcap_{s \in S} \{X^s_{J_s} = a_s\}.$$

Define

$$T(\theta) = \sum_{s \in S} |J_s|, \qquad R(\theta) = |S|.$$

Theorem 4.2 (Multi-sample shadow descent)**.** *Assume by induction that all ranks $< k$ satisfy the bound $B_j \le D^j$. Let $\mathcal{C}$ be a rank-k balanced two-collision code with uniform law μ, and let $\theta = ((J_s, a_s))_{s \in S}$ be a nonempty terminal shadow value for independent root samples from μ. Let E_θ be an event compatible with θ and suppose it has mass*

$$\delta = \mathbb{P}[E_\theta] \ge \exp(-\lambda T(\theta) - c R(\theta)).$$

If $\log D \ge \lambda + c$, then $|\mathcal{C}| \le D^k$.

Proof. Let $M = |\mathcal{C}|$. For each sample label s, let

$$\eta_s = \mu(\{x : x_{J_s} = a_s\}).$$

Since the samples are independent and the terminal event is contained in the intersection of the cylinders,

$$\delta \le \prod_{s \in S} \eta_s.$$

The cylinder $\{x : x_{J_s} = a_s\}$ is again a balanced two-collision code after the coordinates in J_s are deleted: if three remaining codewords are distinct, the original balanced property gives a $2+1$ coordinate, and it cannot be one of the fixed coordinates. Hence the cylinder has rank at most $k - |J_s|$, and

$$\eta_s M \le D^{k-|J_s|}.$$

Multiplying over s gives

$$M^{R(\theta)} \le \delta^{-1} D^{R(\theta)k - T(\theta)} \le \exp(\lambda T(\theta) + cR(\theta)) D^{R(\theta)k - T(\theta)}.$$

Since each nonempty sample label contributes at least one pinned coordinate, $T(\theta) \ge R(\theta)$. If $\log D \ge \lambda + c$, the exponential factor is at most $D^{T(\theta)}$, and so $M^{R(\theta)} \le D^{R(\theta)k}$. Because $R(\theta) \ge 1$, taking $R(\theta)$-th roots gives $M \le D^k$. □

5 The Shadow Entropy Criterion

The proof tree should not count full leaves. It should merge leaves with the same projected shadow and count only the resulting random variable.

Theorem 5.1 (Shadow entropy criterion)**.** *Let Θ be the terminal projected shadow after all analysis-only data is marginalized, and suppose every retained terminal outcome has a nonempty shadow. Suppose*

$$\mathbf{H}(\Theta) \le \mathbb{E}[\lambda T(\Theta) + cR(\Theta)].$$

Then some shadow value θ satisfies

$$-\log \mathbb{P}[\Theta = \theta] \le \lambda T(\theta) + cR(\theta),$$

and Theorem 4.2 applies.

Proof. For $I(\Theta) = -\log \mathbb{P}[\Theta]$ one has $\mathbb{E}I(\Theta) = \mathbf{H}(\Theta)$. Hence

$$\mathbb{E}[I(\Theta) - \lambda T(\Theta) - cR(\Theta)] \le 0.$$

At least one atom has nonpositive value of the random variable inside the expectation. □

We shall use a potential version.

Theorem 5.2 (Stateful shadow entropy with potential)**.** *Let a parent public state S have rank k and potential Φ_{par}. Along a retained terminal branch let $\Delta\Phi$ be the nonnegative potential expenditure charged by the proof tree after all hash tokens, birth deposits, old queues, and scale charges on that branch have been assigned, and let L_{pr} be the accumulated charged pruning-normalization loss of Proposition 3.10. Suppose every retained terminal outcome has a nonempty projected shadow. If*

$$\mathbf{H}(\Theta) + \mathbb{E}L_{\mathrm{pr}} \le \mathbb{E}[\lambda T(\Theta) + cR(\Theta) + A\Delta\Phi],$$

then some retained terminal outcome gives a shadow atom satisfying the corresponding pointwise inequality with respect to its original pre-pruning mass. For a shadow value

$$\theta = ((J_s, a_s))_{s\in S},$$

let $\mathsf{S}_{\theta,s}$ be the cylinder child state obtained from the terminal public ledgers on the branch by restricting the s-th root sample to $x_{J_s} = a_s$ and deleting those pinned coordinates from the current ambient coordinate set, all active supports, old supports, queues, and incidence counters. The underlying code of $\mathsf{S}_{\theta,s}$ is the full parent value cylinder $\{x \in \mathcal{C} : x_{J_s} = a_s\}$ with J_s deleted; non-value analysis atoms from the terminal leaf are inherited only as ledger/support data and do not further shrink this cylinder. Assume the charged branch and the inherited cylinder children satisfy the recurrence-compatibility inequality

$$\Delta\Phi + \sum_{s\in S} \Phi(\mathsf{S}_{\theta,s}) \le R(\theta)\Phi_{\mathrm{par}}.$$

If the induction hypothesis is strengthened to

$$B(\mathsf{S}) \le D^{\mathrm{rank}(\mathsf{S})} \exp(A\Phi(\mathsf{S})),$$

and $\log D \ge \lambda + c$, then the potential term is compatible with the multi-shadow descent.

Proof. Set

$$I_{\mathrm{orig}}(\Theta) = -\log \mathbb{P}_{\mathrm{orig}}[\Theta],$$

where $\mathbb{P}_{\mathrm{orig}}$ is the retained shadow mass in the original pre-pruning probability space. By Proposition 3.10,

$$\mathbb{E}I_{\mathrm{orig}}(\Theta) \le \mathbf{H}(\Theta) + \mathbb{E}L_{\mathrm{pr}}.$$

Hence

$$\mathbb{E}[I_{\mathrm{orig}}(\Theta) - \lambda T(\Theta) - cR(\Theta) - A\Delta\Phi] \le 0,$$

so some retained terminal outcome ω with shadow $\theta = \Theta(\omega)$ satisfies

$$-\log \mathbb{P}_{\mathrm{orig}}[\Theta = \theta] \le \lambda T(\theta) + cR(\theta) + A\Delta\Phi(\omega).$$

For such an atom the shadow descent proof is the stateful version of Theorem 4.2. Write $R = |S|$. For each sample label s, the cylinder child $\mathsf{S}_{\theta,s}$ has as underlying code the full value cylinder $\{x : x_{J_s} = a_s\}$, with the terminal public ledgers inherited only as accounts after deleting J_s. Thus the usual cylinder probability bound is unchanged, and the child is a balanced two-collision code of rank at most $k - |J_s|$. The induction hypothesis gives

$$|\mathcal{C}_{\theta,s}| \le D^{k-|J_s|} \exp(A\Phi(\mathsf{S}_{\theta,s})).$$

Multiplying the usual cylinder estimates gives the same inequality as in Theorem 4.2, but with the additional factor

$$\exp\left(A\Delta\Phi + A\sum_{s\in S} \Phi(\mathsf{S}_{\theta,s})\right).$$

By the recurrence-compatibility condition,

$$\Delta\Phi + \sum_{s\in S} \Phi(\mathsf{S}_{\theta,s}) \le R\Phi_{\mathrm{par}}.$$

Hence, after taking R-th roots, the strengthened induction closes with the parent factor $\exp(A\Phi_{\mathrm{par}})$. The child in this paragraph is the cylinder state carrying the terminal public ledgers; it is not an intermediate transition child and it is not a freshly reset root state.

The drop $\Delta\Phi$ and the inherited child potentials need not be functions of the projected shadow. The averaging argument is applied on the underlying terminal outcome space: it produces an outcome ω with shadow $\theta = \Theta(\omega)$ and the displayed inequality using the branch ledgers of that outcome. The event $\{\Theta = \theta\}$ is still contained in the same value cylinders defining the full cylinder children above, because all analysis-only data has been projected away. Therefore replacing the full leaf by the merged shadow only increases the compatible cylinder event mass and does not change the child code sets used in the induction. □

6 Elementary Lemmas Used Throughout

This section records the elementary tools which appear repeatedly in the argument. We use the standard discrete entropy and relative-entropy conventions from information theory [6, 8, 21]; the order-m Renyi entropy used in copying lemmas is the classical one [20].

Lemma 6.1 (Entropy atom lemma)**.** *Let Y be a finite-valued random variable. There is a value y with positive mass such that*

$$-\log \mathbb{P}[Y = y] \leq \mathbf{H}(Y).$$

More generally, for every $b > 1$, the set of atoms satisfying

$$-\log \mathbb{P}[Y = y] \leq b\mathbf{H}(Y)$$

has total mass at least $1 - 1/b$.

Proof. The first claim follows by averaging the information random variable $-\log \mathbb{P}[Y]$. The second follows from Markov's inequality:

$$\mathbb{P}[-\log \mathbb{P}[Y] > b\mathbf{H}(Y)] \leq 1/b.$$

□

Lemma 6.2 (Collision controls split coordinates)**.** *Let ν be a probability law on a code $\mathcal{C}$, and let X, Y, Z be independent samples from ν. For a coordinate i write*

$$\mathrm{Coll}_i(\nu) = \mathbb{P}[X_i = Y_i].$$

Then

$$\mathbb{P}[i \in \mathrm{Split}(X, Y, Z)] \leq 3\,\mathrm{Coll}_i(\nu),$$

where $i \in \mathrm{Split}(X, Y, Z)$ means that $\{X_i, Y_i, Z_i\}$ has size exactly 2. Consequently, for every block I,

$$\mathbb{E}|\,\mathrm{Split}_I(X, Y, Z)| \leq 3 \sum_{i \in I} \mathrm{Coll}_i(\nu).$$

If the average collision on I is at most α, then

$$\mathbb{P}[|\,\mathrm{Split}_I(X, Y, Z)| \leq 6\alpha|I|] \geq 1/2.$$

Proof. Write $p_a = \mathbb{P}[X_i = a]$. The probability of a $2+1$ split at coordinate i is

$$3\sum_a p_a^2(1-p_a) \le 3\sum_a p_a^2 = 3\,\mathrm{Coll}_i(\nu).$$

Summing over $i \in I$ gives the expectation bound, and Markov's inequality gives the final statement. □

Lemma 6.3 (Exact sparse split exposure). *Let $W = \mathrm{Split}_I(X, Y, Z)$ and suppose we condition on the event $|W| \le s$. Then some set $D \subseteq I$, $|D| \le s$, has conditional mass at least*

$$\left(\sum_{r\le s}\binom{|I|}{r}\right)^{-1}.$$

On the atom $W = D$, the triple is compatible on $I \setminus D$.

Proof. Under $|W| \le s$, the random set W has at most $\sum_{r\le s}\binom{|I|}{r}$ possible values. Choose a most likely atom. The compatibility assertion is the definition of W. □

Lemma 6.4 (Role refinement). *Suppose a triple is split on every coordinate of a block O. Then after conditioning on a role vector of mass at least $3^{-|O|}$, there is a subblock $O' \subseteq O$ with $|O'| \ge |O|/3$ and one ordered pair, say (X, Y) after relabelling, such that*

$$X_{O'} = Y_{O'}, \qquad Z_i \ne X_i \quad (i \in O').$$

The role-exposure cost is $O(|O'|)$.

Proof. There are at most $3^{|O|}$ role vectors. On a fixed role vector, one of the three roles occurs on at least one third of the coordinates. This majority class is O'. □

Lemma 6.5 (KL chain rule and public coordinate localization). *Let $A = (A_1, \dots, A_m)$ have laws $P \ll Q$. For a prefix $u = A_{<t}$ define*

$$d_t(u) = D(P(A_t \in \cdot \mid A_{<t} = u) \| Q(A_t \in \cdot \mid A_{<t} = u)).$$

Then

$$D(P\|Q) = \sum_{t=1}^{m} \mathbb{E}_{u\sim P(A_{<t})} d_t(u).$$

If a coordinate is chosen by the first t satisfying a deterministic threshold in a fixed public order, then the coordinate identity is public. The only random selector is the prefix value u.

Proof. The KL identity is the standard chain rule for relative entropy [6, 8, 15], here written from the product rule for conditional probabilities:

$$\log\frac{P(A)}{Q(A)} = \sum_t \log\frac{P(A_t \mid A_{<t})}{Q(A_t \mid A_{<t})}.$$

Taking expectation under P gives the identity. The publicness statement is by construction. □

Lemma 6.6 (Value-pin or hash dichotomy)**.** *Let p be a law on a finite alphabet and S a set with $p(S) \geq \rho > 0$. Let p_S be the conditional law on S. For $0 < \tau \leq 1$, either some $a \in S$ satisfies*

$$p(a) \geq \rho\tau,$$

or

$$\mathrm{Coll}(p_S) = \sum_a p_S(a)^2 \leq \tau.$$

Proof. If $\max_{a \in S} p_S(a) \geq \tau$, then the first alternative holds. Otherwise

$$\mathrm{Coll}(p_S) \leq \max_{a \in S} p_S(a) \sum_{a \in S} p_S(a) < \tau.$$

□

Lemma 6.7 (Heavy atoms or low-collision complement)**.** *Let p be a law on a finite alphabet and fix $0 < \rho \leq 1/2$ and $0 < \tau < 1$. Put*

$$H = \{a : p(a) \geq \rho\tau\}, \qquad L = \Omega \setminus H.$$

Then $|H| \leq (\rho\tau)^{-1}$, every branch $V = a \in H$ is a terminal value pin of conditional mass at least $\rho\tau$, and exactly one of the following holds for the light complement:

(i) *$p(L) < \rho$, in which case pruning L costs at most 2ρ;*

(ii) *$p(L) \geq \rho$, in which case the conditional law p_L has $\mathrm{Coll}(p_L) \leq \tau$.*

Proof. The cardinality bound on H is immediate. If $p(L) < \rho$, the pruning cost is $-\log(1-p(L)) \leq 2\rho$. If $p(L) \geq \rho$, then every $a \in L$ has $p(a) < \rho\tau$, so

$$p_L(a) = \frac{p(a)}{p(L)} < \tau.$$

Therefore

$$\mathrm{Coll}(p_L) \leq \max_a p_L(a) \sum_a p_L(a) < \tau.$$

□

Lemma 6.8 (Positive KL tail)**.** *Let $p \ll q$ and $D(p\|q) > 0$. Then*

$$S_+ = \{a : p(a) > q(a)\}$$

has positive p-mass. If $p(S_+) \geq \rho$, Lemma 6.6 applies on S_+. If $p(S_+) < \rho$, pruning S_+ costs at most $-\log(1 - p(S_+)) \leq 2\rho$ for $\rho \leq 1/2$, after which Lemma 6.6 applies on the complement.

Proof. If $p(S_+) = 0$, then $\log(p/q) \leq 0$ p-almost surely, contradicting positive KL. The pruning cost follows from $-\log(1 - x) \leq 2x$ for $0 \leq x \leq 1/2$. □

In the transition system, this positive-tail pruning is not an independent operation which may be repeated for free. The retained complement $V_t \notin S_+$ is appended to the same public prefix/tail transcript as the subsequent value pin or hash coordinate, and its charge $-\log(1 - p(S_+))$ is included in that local terminal/hash charge. The discarded branch $V_t \in S_+$ is a charged pruning branch and is not inherited by any cylinder child. If this retained complement would exceed the remaining fixed-window budget, the pruning step is not taken on the continuing branch: the branch terminalizes on the already paid coordinate-value prefix before any set-valued complement event is appended.

Lemma 6.9 (One-coordinate collision exits are exhaustive tuple exits)**.** *Let p, q be one-coordinate laws and let $0 < \rho \leq 1/2$. If*

$$\sum_a p(a)q(a) > \gamma_0,$$

then either $\max_a p(a) > \gamma_0^2$ *or* $\max_a q(a) > \gamma_0^2$*. If*

$$\mathrm{Coll}(p) = \sum_a p(a)^2 > \alpha,$$

then $\max_a p(a) > \alpha$*. Consequently, whenever $\rho\tau \leq \min(\alpha, \gamma_0^2)$, the cross-collision and high-side-collision exits in Theorem 12.4 route exhaustively through one actual sample-coordinate law r: heavy atom branches of r are terminal projected tuple pins of conditional mass at least $\rho\tau$, a complement of mass $< \rho$ is a charged pruning branch, and a complement of mass at least ρ gives a public low-collision hash coordinate under the conditional law on that complement.*

Proof. By Cauchy's inequality,

$$\sum_a p(a)q(a) \leq \sqrt{\mathrm{Coll}(p)\,\mathrm{Coll}(q)}.$$

If both maxima were at most γ_0^2, then both collisions would be at most γ_0^2, contradicting the strict cross-collision inequality. The high-side statement follows from $\mathrm{Coll}(p) \leq \max_a p(a)$. In the cross-collision case choose $r = p$ or $r = q$ according to the fixed public tie-break among the laws with a large atom; in the high-side case choose $r = p$. Since $\max_a r(a) > \min(\alpha, \gamma_0^2) \geq \rho\tau$, Lemma 6.7 applies to r. Its heavy branches are projected tuple pins by Lemma 8.2; its small complement is charged pruning; and its large complement has low collision for the actual conditional sample-coordinate law after the public complement event is paid in the stopped hash transcript. □

7 Closed Auxiliary Results

This section records closed tools which are useful independently of the final global normal-form assembly.

Lemma 7.1 (Pair codegree in a 3-uniform sunflower-free family)**.** *Let $\mathcal{F}$ be a 3-uniform family with no three-petal sunflower. Then every pair of vertices is contained in at most two members of $\mathcal{F}$.*

Proof. If three distinct triples contain the same pair $\{u, v\}$, then their pairwise intersections are all exactly $\{u, v\}$, so they form a sunflower with kernel $\{u, v\}$. □

Lemma 7.2 (Vertex links have at most six edges)**.** *Let $\mathcal{F}$ be a 3-uniform family with no three-petal sunflower. For every vertex v, the link graph L_v has maximum degree at most 2, matching number at most 2, and at most 6 edges. Hence $\deg_{\mathcal{F}}(v) \leq 6$.*

Proof. If a vertex u has degree at least 3 in L_v, then three triples contain the pair $\{u, v\}$, contradicting Lemma 7.1. Thus $\Delta(L_v) \leq 2$. If L_v contains a matching of size 3, then the corresponding three triples in $\mathcal{F}$ all meet pairwise exactly in v, forming a sunflower with kernel $\{v\}$. Hence $\nu(L_v) \leq 2$.

A graph of maximum degree at most 2 is a disjoint union of paths and cycles. If it has more than 6 edges, then its components contain a matching of size at least 3: each path or cycle with e edges has matching number at least $\lfloor e/2 \rfloor$, and distributing at most two matching edges among all components permits at most six edges, with equality only for two triangles. Therefore $|E(L_v)| \leq 6$. □

Theorem 7.3 (Pairwise-intersecting 3-uniform local theorem)**.** *Let $\mathcal{F}$ be a pairwise-intersecting 3-uniform family with no three-petal sunflower. Then*

$$|\mathcal{F}| \le 10.$$

If equality holds, then, up to relabelling, $\mathcal{F}$ is the 2-$(6, 3, 2)$ *design.*

Proof. The pair-codegree bound is Lemma 7.1. We first record a small graph fact. If two graphs X, Y have maximum degree at most 2 and every edge of X intersects every edge of Y, then $|X| + |Y| \le 6$. Indeed, any edge of Y is a two-vertex cover of X, and conversely. A graph of maximum degree at most 2 with a two-vertex cover has at most 4 edges; if one side has 4 edges, its possible two-covers form at most two edges for the other side, while if one side has 3 edges the possible two-covers form at most three edges. The cases with at most two edges are immediate.

Let $\tau(\mathcal{F})$ be the vertex-covering number. Since any member of $\mathcal{F}$ meets all other members, $\tau(\mathcal{F}) \le 3$. If $\tau(\mathcal{F}) \le 2$, fix a two-vertex cover $\{1, 2\}$. Apart from the at most two triples containing both 1 and 2, the triples through 1 and the triples through 2 give two cross-intersecting link graphs of maximum degree at most 2. The graph fact gives $|\mathcal{F}| \le 2 + 6 = 8$.

It remains to treat $\tau(\mathcal{F}) = 3$. Fix a triple $T = \{1, 2, 3\} \in \mathcal{F}$. Every other member meets T. For each pair $ij \subset T$, let H_{ij} be the possible extra triple containing that pair. By Lemma 7.1 there is at most one such extra triple for each pair. Let h be the number of existing H_{ij}.

For $i = 1, 2, 3$, let X_i be the graph of outside pairs e for which

$$\{i\} \cup e \in \mathcal{F}$$

and this triple contains no other vertex of T. Each X_i is an intersecting graph of maximum degree at most 2: two disjoint edges in X_i, together with the edge $T \setminus \{i\}$ in the link L_i, would give a matching of size 3 in L_i and hence a sunflower with kernel $\{i\}$. Therefore $|X_i| \le 3$, with equality only for a triangle. Also X_i and X_j are cross-intersecting for $i \ne j$, because $\mathcal{F}$ is pairwise intersecting.

If $H_{12} = \{1, 2, a\}$ exists, then every edge of X_3 contains a, since the corresponding triple must meet H_{12} outside T. Hence $|X_3| \le 2$. Moreover $\deg_{X_1}(a) \le 1$ and $\deg_{X_2}(a) \le 1$, because the pair $\{1, a\}$ or $\{2, a\}$ already occurs in H_{12} and cannot occur in two further triples.

The capacity bounds now give

h	$\lvert\mathcal{F}\rvert$
0	$\le 1 + 8$
1	$\le 1 + 1 + (3 + 3 + 2)$
2	$\le 1 + 2 + (3 + 2 + 2)$
3	$\le 1 + 3 + (2 + 2 + 2).$

For $h = 0$, the value 1+9 would require three pairwise cross-intersecting triangles. Cross-intersecting triangles must have the same vertex set, which would make every outside pair occur in three triples, violating Lemma 7.1. Thus the $h = 0$ case gives at most 9. All cases give $|\mathcal{F}| \le 10$.

We finally identify equality. If $h = 1$, say $H_{12} = \{1, 2, a\}$, equality would force X_1 and X_2 to be cross-intersecting triangles, hence the same triangle. The two edges of X_3 must hit that triangle and also contain a, forcing an outside pair to appear too often. Thus $h = 1$ cannot give equality.

If $h = 2$, say $H_{12} = \{1, 2, a\}$ and $H_{13} = \{1, 3, b\}$, equality would force X_1 to be a triangle and X_2, X_3 to be two-edge stars. The forced-star condition makes X_2 centered at b, and the cross-intersection with the triangle X_1 forces $\deg_{X_1}(b) = 2$, contradicting the degree cap from H_{13}.

Thus equality has $h = 3$. Write

$$H_{12} = \{1, 2, a\}, \quad H_{13} = \{1, 3, b\}, \quad H_{23} = \{2, 3, c\}.$$

Then X_1, X_2, X_3 are two-edge stars centered respectively at c, b, a. The degree caps force a, b, c to be distinct. Two cross-intersecting two-edge stars with different centers must share the center-edge and have their remaining edges point to the same third vertex. Applying this to the three pairs of stars gives, after relabelling,

$$X_1 = \{ac, bc\}, \qquad X_2 = \{ab, bc\}, \qquad X_3 = \{ab, ac\}.$$

Hence the ten triples are

$$123,\ 12a,\ 13b,\ 23c,\ 1ac,\ 1bc,\ 2ab,\ 2bc,\ 3ab,\ 3ac.$$

Every pair of the six vertices $\{1, 2, 3, a, b, c\}$ occurs in exactly two triples, so this is the 2-$(6, 3, 2)$ design. The preceding forced choices also prove uniqueness. □

Corollary 7.4 (Size-nine stability in the pairwise case)**.** *Let $\mathcal{F}$ be a pairwise-intersecting 3-uniform family with no three-petal sunflower and $|\mathcal{F}| = 9$. Then $\mathcal{F}$ is obtained from the* 2-$(6, 3, 2)$ *design by deleting one triple.*

Proof. Use the notation from the proof of Theorem 7.3. The case $\tau(\mathcal{F}) \le 2$ gives at most 8 triples, so $\tau(\mathcal{F}) = 3$. The case $h = 0$ would require sizes $(3, 3, 2)$ among X_1, X_2, X_3; the two triangles must coincide, and the third graph must use edges of the same triangle, producing pair-codegree 3. Thus $h = 0$ is impossible. If $h = 1$, then either two of the X_i are coincident triangles, giving the same pair-codegree violation, or one triangle is crossed by a forced two-edge star, forcing degree 2 at the special vertex and contradicting the degree cap from the corresponding H_{ij}.

For $h = 2$, say H_{12} and H_{13} exist, the capacity bound forces all three graphs X_1, X_2, X_3 to have two edges. The forced centers and the cross-intersecting two-star calculation used in Theorem 7.3 determine the unique configuration, namely the 2-$(6, 3, 2)$ design with the missing triple H_{23} deleted. For $h = 3$, the sizes of X_1, X_2, X_3 are $(2, 2, 1)$ up to permutation; the same two-star calculation determines the unique configuration, now the 2-$(6, 3, 2)$ design with one X_i-type triple deleted. These are exactly the one-triple deletions of the design. □

Proposition 7.5 (Linear rank case)**.** *Let $\mathcal{F}$ be a rank at most k linear family with no three-petal sunflower. Then*

$$|\mathcal{F}| \le 2k + 2.$$

Proof. Linearity implies that three sets through one vertex would form a sunflower with that vertex as kernel, so every vertex degree is at most 2. The family also has matching number at most 2, because three disjoint sets form a sunflower with empty kernel.

Let $M = |\mathcal{F}|$ and form the disjointness graph G on vertex set $\mathcal{F}$. Since the matching number of $\mathcal{F}$ is at most 2, the graph G is triangle-free. For any $A \in \mathcal{F}$, at most k other members of $\mathcal{F}$ intersect A: each vertex of A is contained in at most one further member. Hence

$$\deg_G(A) \ge M - 1 - k.$$

Thus

$$|E(G)| \ge \frac{M(M - 1 - k)}{2}.$$

By Mantel's theorem [16], $|E(G)| \le M^2/4$. For $M > 0$ this gives $M - 1 - k \le M/2$, hence $M \le 2k + 2$. □

Theorem 7.6 (Projection-profile slice-rank bound)**.** *Let $\mathcal{C} \subseteq \prod_{i=1}^{k} \Omega_i$ be a balanced two-collision code. For a map*

$$\sigma : [k] \to \{0, xy, xz, yz\}$$

define $R_x(\sigma)$ to be the coordinates where $\sigma(i)$ involves x, and define $R_y(\sigma), R_z(\sigma)$ similarly. Then

$$|\mathcal{C}| \leq \sum_{\sigma} \min_{u \in \{x,y,z\}} |\pi_{R_u(\sigma)}(\mathcal{C})|.$$

Proof. This is a projection-profile variant of the slice-rank polynomial method [7,9,22]. Work over $\mathbb{F}_3$. On one coordinate define

$$h(a, b, c) = 1 - \mathbf{1}_{a=b} - \mathbf{1}_{a=c} - \mathbf{1}_{b=c}.$$

In $\mathbb{F}_3$ this equals 0 exactly when exactly two of a, b, c are equal, and equals 1 when all three are equal or all three are distinct. Set

$$H(x, y, z) = \prod_{i=1}^{k} h(x_i, y_i, z_i).$$

Restricted to $\mathcal{C}^3$, the balanced two-collision property gives

$$H(x, y, z) = \begin{cases} 1, & x = y = z, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the slice rank of $H|_{\mathcal{C}^3}$ is $|\mathcal{C}|$.

Expanding the product, each term is indexed by $\sigma : [k] \to \{0, xy, xz, yz\}$. For such a term, the x-dependence is only through $\pi_{R_x(\sigma)}(x)$, so it has slice rank at most $|\pi_{R_x(\sigma)}(\mathcal{C})|$ when sliced in the x variable. The same argument applies to y and z. Hence the term has slice rank at most the minimum of the three displayed projection sizes. Slice rank is subadditive, giving the claim. □

Corollary 7.7 (Bounded-alphabet and variable-profile bounds)**.** *If $|\Omega_i| \leq Q$ for all i, then every balanced two-collision code satisfies*

$$|\mathcal{C}| \leq (1 + 3Q^{2/3})^k \leq (4Q^{2/3})^k.$$

More generally, if every projection satisfies $|\pi_R(\mathcal{C})| \leq D^{|R|}$, then

$$|\mathcal{C}| \leq (1 + 3D^{2/3})^k.$$

Proof. From Theorem 7.6,

$$|\mathcal{C}| \leq \sum_{\sigma} \min_{u} D^{|R_u(\sigma)|} \leq \sum_{\sigma} D^{(|R_x(\sigma)|+|R_y(\sigma)|+|R_z(\sigma)|)/3}.$$

For each coordinate, the choice 0 contributes 0 to the sum $|R_x| + |R_y| + |R_z|$, while each of xy, xz, yz contributes 2. Therefore the last sum equals $(1+3D^{2/3})^k$. The bounded-alphabet statement follows by taking $D = Q$. □

8 Tuple Nodes and Conditioned Copying

Local close-pair and hash statements are often true only under a tuple-level conditioning event. They must not be converted directly into code-level statements.

Definition 8.1 (Tuple node)**.** *A tuple node is an event*

$$E \subseteq \mathcal{C}^r$$

for independent samples $X^1, \dots, X^r$. All local laws at the node are conditional laws under $\mathbb{P}(\cdot \mid E)$.

Lemma 8.2 (Tuple-level pins are projected shadows)**.** *At a tuple node $E \subseteq \mathcal{C}^r$, suppose a branch terminalizes by fixing $X^s_J = a$ for one of the active sample labels s. Then the terminal event is compatible with the projected shadow obtained by appending (J, a) to the s-sample list. The mass of the tuple node and of the local atom is counted by the proof-tree entropy ledger; no ambient-code density increment is being asserted.*

Proof. The terminal event is a subset of

$$E \cap \{X^s_J = a\} \subseteq \{X^s_J = a\}.$$

This is exactly the compatibility condition in the definition of a multi-sample value-shadow. The probability of reaching E and then choosing the local atom is part of the full leaf transcript Γ and is therefore accounted for by Proposition 3.9. After projecting away the analysis transcript, only the cylinder $\{X^s_J = a\}$ remains in the projected shadow. □

Lemma 8.3 (Tuple prefixes project to multi-sample shadows)**.** *At a tuple node $E \subseteq \mathcal{C}^r$, suppose a branch terminalizes by fixing a finite public list of tuple-coordinate values*

$$X^{s_\ell}_{j_\ell} = a_\ell \qquad (\ell = 1, \dots, L),$$

with sample labels and coordinates determined by the paid public prefix. First discard from this list every exact repeat whose value is already forced by an earlier retained tuple-coordinate value; such a repeat has zero conditional entropy in the prefix chain. If an exact repeat asks for a different value, the branch has zero mass and is ignored. For every remaining paid variable choose one sample-label coordinate pin, grouping the chosen pins by sample label. Then the number of entropy-bearing prefix variables is at most $T(\Theta)$ for the resulting projected shadow. The terminal event is compatible with this multi-sample projected value-shadow, and its local probability cost is the entropy of the paid prefix list in the proof-tree ledger, not an ambient-code density increment.

Proof. After the zero-entropy repeats are discarded, the terminal event is contained in the intersection of the retained coordinate cylinders. For a tuple-coordinate variable which is already a common value of two labels, choosing either label gives a cylinder containing the terminal event. Grouping the chosen pins by sample label rewrites this intersection in the form required by Definition 4.1. The full prefix values, including the deterministic repeats before projection, remain in the terminal leaf transcript before projection and are paid by Proposition 3.9. □

Lemma 8.4 (Fixed-node conditioned copying)**.** *Let $E \subseteq \mathcal{C}^r$ have mass $\delta > 0$, and let*

$$\nu = \mathrm{Law}(X^s \mid E).$$

If $F \subseteq \mathcal{C}^m$ satisfies $\nu^m(F) \geq \rho$, then in m independent copies of the whole tuple experiment the event that all copies lie in E and their selected s-samples satisfy F has mass at least $\delta^m \rho$.

Proof. Conditioning the m independent copies on E^m makes the selected samples independent with common law ν. The mass is exactly $\delta^m \nu^m(F)$. □

Lemma 8.5 (Random-transcript conditioned copying). *Let K be a finite public key whose support has already been restricted to the surviving key atoms, with atoms k and probabilities p_k. On the atom $K = k$, let ν_k be the selected one-sample conditional law. Suppose that for every atom in this support there is an event F_k for m independent draws from ν_k satisfying*

$$\nu_k^m(F_k) \geq \rho.$$

Then in m independent copies of the whole stopped experiment, the event that all copies have the same key k and their selected samples satisfy F_k has mass at least

$$\rho \sum_k p_k^m = \rho \exp(-(m-1)\mathbf{H}_m(K)),$$

where

$$\mathbf{H}_m(K) = \frac{1}{1-m} \log \sum_k p_k^m$$

is the order-m Renyi entropy.

Proof. The desired mass is

$$\sum_k p_k^m \nu_k^m(F_k).$$

Since the support of K consists only of surviving atoms, the uniform lower bound $\nu_k^m(F_k) \geq \rho$ applies to every summand and gives the displayed estimate. In applications one first conditions the transcript on the good-key event and pays the fixed mass loss separately. □

Thus for random transcripts the copy cost is the Renyi entropy of the public key that is shared by the copies. If K is the public key and $m = 3$, the copy tax is

$$2\mathbf{H}_3(K) + O(1).$$

Thus hash returns must use public keys whose entropy is already paid.

9 Public Hash Windows

The following theorem is the corrected form of the fixed hash return.

Definition 9.1 (Labelled hash-window survival convention). *Fix $0 < \tau \leq \beta < 1/12$. A hash window also fixes a public active sample label s. Every produced hash coordinate is an ambient coordinate of X^s under the current paid tuple event; coordinates belonging to different sample labels are placed in different hash windows. A produced hash coordinate is live after a transcript prefix if its current one-coordinate conditional law has no atom of mass $\geq \beta$. The hash-window rule scans the public list of produced coordinates after every transcript extension. If a produced coordinate is not live, the proof chooses the first public atom of mass at least β, restricts to that atom, and stops as a terminal value-pin shadow; the discarded complement is a charged pruning loss at most $\log(1/\beta)$ and is not inherited by any child. A coordinate which is live at the terminal transcript is called surviving. Since collision is bounded by the largest atom, every surviving coordinate has terminal collision at most β. Since a codeword atom is bounded by any of its coordinate atoms on a nonempty surviving block, the conditional one-sample law on a surviving hash block has maximum codeword atom $< \beta$. In a fixed-window statement,* Destroyed *denotes the number of produced coordinates which are not surviving, after terminal value-pin and bounded-support exits have been removed.*

Theorem 9.2 (Public-transcript fixed-window hash return)**.** *Fix a public state P_0. Run a bounded hash window with terminal transcript T satisfying $\mathbf{H}(T \mid P_0) \le B_{\mathrm{win}}$. Suppose coordinate production is deterministic from P_0 and the transcript prefix, and suppose all produced coordinates are attached to the same public sample label s. Suppose every non-bounded branch produces at least $2h$ distinct hash coordinates, and the adaptive destruction budget satisfies*

$$\mathbb{E}[\mathrm{Destroyed} \mid P_0] \le h/2.$$

Then, except for terminal value-pin or bounded-support exits, there is a transcript event of conditional mass at least $1/2$ on which a block $H(T)$ of h surviving hash coordinates is deterministic from the paid transcript. For each transcript in that event, three conditioned copies of the conditional law of the same sample label X^s are distinct and give a near-sunflower event on $H(T)$ with conditional probability at least $\rho_{\mathrm{hash}} := 1/2 - 3\beta > 0$. Equivalently, the returned near-sunflower has a fixed positive mass, and its total proof-tree cost is

$$O(B_{\mathrm{win}} + h).$$

Here the cost is the local entropy/mass charge used in Proposition 3.9: the public transcript entropy, the order-three Renyi copy tax for the transcript key, and the exact sparse split exposure on the returned block.

Proof. If fewer than h coordinates survive, more than h are destroyed. Markov's inequality gives probability at most $1/2$ for this bad event. On the good event choose the first h surviving produced coordinates. This block is a deterministic function of (P_0, T), so it carries no entropy beyond the paid transcript.

Let G be the good-transcript event and let T_G be T conditioned on G. Since $\mathbb{P}(G \mid P_0) \ge 1/2$ and $\mathbf{H}(T \mid P_0) \le B_{\mathrm{win}}$,

$$\mathbf{H}_3(T_G \mid P_0) \le \mathbf{H}(T_G \mid P_0) \le 2B_{\mathrm{win}} + O(1).$$

For each good transcript t, let

$$\nu_t = \mathrm{Law}(X^s \mid P_0, T = t).$$

Every coordinate of $H(t)$ has collision at most β under the corresponding one-coordinate marginal of ν_t. Hence Lemma 6.2 gives the near-sunflower split event for three independent samples from ν_t with probability at least $1/2$:

$$\nu_t^3\{|\mathrm{Split}_{H(t)}(X, Y, Z)| \le 6\beta h\} \ge 1/2.$$

Because $H(t)$ is nonempty and all its coordinates are surviving,

$$\max_x \nu_t(x) < \beta.$$

Thus the probability that three independent ν_t-samples are not all distinct is at most

$$3 \sum_x \nu_t(x)^2 \le 3\beta.$$

With $\beta < 1/12$, the distinct near-sunflower event

$$F_t = \{X, Y, Z \text{ distinct and } |\mathrm{Split}_{H(t)}(X, Y, Z)| \le 6\beta h\}$$

has $\nu_t^3(F_t) \ge \rho_{\mathrm{hash}} := 1/2 - 3\beta > 0$ for every good transcript t.

Lemma 8.5 realizes these three conditional samples in the original tuple experiment using the common good transcript key $K = T_G$. Since $\mathbb{P}(G \mid P_0) \geq 1/2$, passing from the conditioned key T_G back to the original stopped experiment loses only the fixed factor $\mathbb{P}(G \mid P_0)^3 \geq 2^{-3}$, absorbed in the $O(1)$ term. The copy charge is therefore

$$2\mathbf{H}_3(T_G \mid P_0) + O(1) = O(B_{\mathrm{win}} + 1).$$

Finally the exact split set on $H(T)$ has at most 2^h possible values, and under the sparse threshold used later at most $\sum_{r \leq 6\beta h} \binom{h}{r}$ values; in either case the fixed-window exposure costs $O(h)$. Thus the total charged cost is $O(B_{\mathrm{win}} + h)$. $\square$

Remark 9.3. *This theorem is only a fixed-window theorem. It is designed for a constant-base result, not for optimizing constants. In the normal-form transition system it is applied in the stronger regime* Destroyed $= 0$*, because the survival scan terminalizes as soon as a produced coordinate becomes non-live. The displayed destruction-budget hypothesis is included to isolate the robust fixed-window statement from its particular use in Proposition 9.5.*

Remark 9.4 (Worst-case transcript budget)**.** *The bound $\mathbf{H}_3(T_G \mid P_0) \leq \mathbf{H}(T_G \mid P_0)$ is deliberately worst-case. No saving from nonuniform transcripts is used anywhere in the global charge. The window rule is stronger than an a posteriori entropy bound: a continuing branch appends prefix, tail/pruning, and production data only while the cumulative public description remains within the fixed budget B_{win}. A rare prefix whose next description would exceed the remaining budget terminalizes as a projected value-shadow before it enters the hash window; a set-valued tail/pruning event is never retained as the terminal shadow. Thus every nonterminal window key has already been paid by a fixed stopped transcript budget, and the Renyi copy tax is bounded by the same constant even in the worst transcript distribution.*

Proposition 9.5 (Fixed-window hash ledger)**.** *Choose fixed hash-window parameters*

$$h,\ \beta,\ \tau,\ B_{\mathrm{win}}$$

with $h \geq 1$ and $0 < \tau \leq \beta < 1/12$, and enforce Definition 9.1. Here B_{win} is a budget for one labelled window. A branch may contain several labelled windows over its lifetime, but each such window has one public sample label, one stopped transcript of size at most B_{win}, and one bounded charge later terminalized or assigned by Proposition 9.7. Then every nonterminal hash window generated by Definition 3.5 either satisfies the hypotheses of Theorem 9.2, or exits to one of:

(i) *a terminal projected value pin;*

(ii) *a bounded hash-support block B with $1 \leq |B| < 2h$ whose fresh part $B \setminus U$ is promoted immediately to U;*

(iii) *a bounded hash-support block $B \subseteq U$ whose incidences are registered immediately in the old-incidence ledger, unless the registration triggers the high-revisit exit.*

Moreover every nonterminal hash-window transcript charge is assigned to a terminal, fresh, old-incidence, scale-descent, or close-pair birth certificate before any child state or descendant hash window is entered from the returned block.

Proof. Hash coordinates are produced by Theorem 12.5 from a public coordinate order after the prefix value has been paid. Each produced coordinate is assigned to its public sample label, and hash windows are maintained separately for different labels. Within one window the cumulative coordinate-production transcript is stopped at the fixed budget B_{win}; any branch whose next prefix would exceed the remaining budget terminalizes on that projected prefix instead of adding the coordinate to the window. Hence within one window the coordinate production rule is deterministic from the public state, the public sample label, and the paid transcript prefix. If a value pin occurs at production, or later by the scan in Definition 9.1, the branch terminalizes with the fixed pruning loss specified there. Thus on a nonterminal branch every produced coordinate that remains in the window is surviving and has terminal collision at most β for the conditional law of that same sample label.

If the cumulative budget closes before $2h$ distinct surviving coordinates are produced and none has yet been retained, the branch has already stopped on the projected prefix described above. If no coordinate has been accepted and the ambient close-pair/residual decision stops supplying a coordinate-producing exit, then no hash window has been opened and no hash token is created; the branch follows that non-hash decision. Otherwise let B be the nonempty set of surviving coordinates already produced. Then $|B| < 2h$. If $B \setminus U$ is nonempty, that public fresh part is promoted immediately to U and the bounded hash transcript is charged to the fresh drop. If $B \subseteq U$, the incidences of B are registered in the old ledger; a counter crossing the revisit cap triggers the high-revisit exit, and otherwise the old potential pays the bounded transcript. Thus the bounded-support exit is absorbed without opening another hash window and without appealing back to this proposition. If at least $2h$ surviving coordinates are produced before the budget closes, the transcript entropy is at most B_{win} by construction, coordinate production is public, and the adaptive destruction budget in Theorem 9.2 is zero. Hence the hypotheses of that theorem hold.

It remains to certify the charge of nonterminal windows. If a bounded support contains a fresh coordinate, promote the whole fresh part to U and charge the fixed window cost to the fresh/old potential drop. If the bounded support is contained in U, register its coordinates as old incidences; repetition above the revisit threshold triggers the high-revisit exit, while below that threshold the incidences are debited in N_{old}. If a surviving hash block returns to residual repair, its window charge is carried as a pending charge of that returned block. The residual node must leave residual mode before a new hash window can be opened from it. Its first nonterminal exit is fresh support promotion, an old incidence/window, a deletion-trapped scale descent, or the birth of a public close-pair block; a bounded leaf or terminal shadow pays the charge directly. The pending charge is a bookkeeping token of size at most C_{hash}: assigning it to the first exit means that this amount is added to that exit's local charge before the branchwise entropy sum is estimated. In the deletion-trapped case the assignment is made at the parent residual transition, before the smaller-scale child is entered; the token is added to the scale ledger for that descent or to the birth certificate of the new active block. The child state therefore carries no hash token from its parent. The token is then removed from the returned block's state, not from the entropy ledger already paid. Thus a hard exit receives only the bounded charge of the hash window which created the block currently being processed, so hash transcripts do not form a separate reusable budget. $\square$

Proposition 9.6 (Hash-coordinate accumulation dichotomy)**.** *During one labelled hash window, the proof does not assume that the product-KL or collision exit remains available a prescribed number of times. The window is run with the following exit dichotomy. Each time a one-coordinate exit produces a public low-collision coordinate for the current sample label, its prefix, tail/pruning, and production transcript are appended only if the cumulative paid transcript remains within B_{win}. If this process reaches $2h$ distinct live coordinates, Theorem 9.2 is applied. If the process stops first, then either the branch has already terminalized on a projected value/prefix pin, or the nonempty set of live coordinates accumulated so far is a bounded hash-support block and exits through the fresh/old alternatives of Proposition 9.5.*

Consequently the global proof never needs a lower bound on the frequency of positive-KL triggers inside a window. A window either returns a fixed-size hash near-sunflower, or its already produced bounded support is charged before another hash window can be opened. In particular, no estimate is made or needed for the typical depth of the KL prefix search. Long prefixes are terminal projected value-shadows, where their entropy is paid by the pinned prefix coordinates through $T(\Theta)$; only prefixes whose cumulative description fits inside B_{win} are allowed to contribute to a hash key.

Proof. The coordinate-producing step is exactly the low-collision alternative of Theorem 12.5 or the corresponding high-revisit/collision exit, run under the current paid public transcript. If the step instead gives a value atom or a prefix whose next transcript would exceed the remaining budget, Proposition 9.5 terminalizes it as a projected shadow and no hash coordinate is added. If a coordinate is added, the survival scan in Definition 9.1 immediately terminalizes any later atom of mass at least β, so every nonterminal coordinate in the window is live.

Thus, before the window has produced $2h$ distinct live coordinates, the retained hash state contains only a public set B with $|B| < 2h$ and a paid transcript of length at most B_{win}. If production continues until $2h$ coordinates are present, the hypotheses of Theorem 9.2 hold, with destruction budget zero. If production cannot continue inside the budget, or if the subsequent close-pair/residual decision no longer supplies a coordinate-producing exit for this labelled window, there are two cases. When $B = \varnothing$, no labelled window has yet been opened and no hash transcript or token is retained. When $B \neq \varnothing$, Proposition 9.5 classifies the current bounded set B: its fresh part is promoted, or if $B \subseteq U$ its old incidences are registered unless high revisit triggers. That classification is a terminal exit from the current hash window, so the same bounded transcript cannot be charged again. □

Proposition 9.7 (Hash-token matching). *On each retained branch, the nonterminal fixed-window hash returns which create pending charges admit a matching ϕ from pending hash returns to terminal/fresh/old/scale/birth hard exits. Each return is assigned to the first hard exit supplied by its returned block, and every hard exit is matched to at most one hash return. Consequently every local hard-exit inequality only has to pay one additional C_{hash} token.*

Proof. A pending token is attached to a specific returned residual block. By Proposition 9.5 and the branch invariant in Proposition 3.7, that block is processed until the present token is assigned; before this assignment no second hash-token-bearing window is retained from the block or from a child born inside that block. If the block exits by a terminal shadow, fresh promotion, old-incidence registration, scale descent, or birth of a close-pair block, define ϕ to be that first exit. A birth transition is therefore not a way to pass the token to a descendant: the token is assigned at the parent birth step before the newly born close-pair block is entered. In the deletion-trapped case the assignment is made at the parent residual transition before the smaller-scale child is entered. If a window accepts no coordinate, no token is created, and if it stops as a bounded-support exit, that same fresh/old exit is its image.

Suppose two distinct pending returns were assigned to the same hard exit. The later return would have had to be retained before the earlier token was removed from the same active residual block, or else the hard exit would already have occurred before the later return was created. Both alternatives contradict the first paragraph. Thus ϕ is injective on hard exits. □

Lemma 9.8 (Token-free children). *Every nonterminal child state entered by the transition system is token-free, except for the unique residual state produced immediately by a nonterminal fixed-window hash return. That exceptional residual state carries exactly the token of the window which produced it. Before any child of that residual state is entered, and before any new hash window is* *opened from data produced by that residual state, the token is assigned to the terminal, fresh, old, scale, or birth exit which leaves the residual state.*

Proof. The only transition which creates a pending hash token is the nonterminal return of Theorem 9.2; by Proposition 9.5 the token is attached to the returned residual block. All other hash-window outcomes are terminal value pins or bounded-support exits, and those are charged immediately as terminal, fresh, or old exits.

Consider the next transition out of this returned residual state. The transition system lists only terminal shadow, fresh promotion, old-incidence registration/window, deletion-trapped scale descent, bounded leaf, or birth of a public close-pair block. In the first three nonterminal hard exits the token is included in the local charge of that exit. In a deletion-trapped scale descent it is included in the parent scale step before the smaller-scale child is entered. In a birth transition it is included in the parent birth certificate before the close-pair child is entered. A bounded leaf stops the recurrence and a terminal shadow has no child. Hence every entered child is token-free, and no descendant hash window can be opened while an ancestor token is still pending. □

10 Residual Repair Lemmas

Lemma 10.1 (DRC row block)**.** *Let (Ω, λ) be a probability space, let I be a finite set with $|I| = n$, and let $(E_i)_{i\in I}$ be events satisfying*

$$\frac{1}{n}\sum_{i\in I} \lambda(E_i) \geq \delta$$

for some $0 < \delta \leq 1$. If

$$1 \leq s \leq \delta n/4,$$

then there is a set $J \subseteq I$, $|J| = s$, such that

$$\lambda\Big(\bigcap_{j\in J} E_j\Big) \geq \frac{\delta}{2}\left(\frac{\delta}{4}\right)^s.$$

For fixed δ, conditioning on this DRC atom costs $O_\delta(s)$ in the proof-tree ledger.

Proof. This is the one-row form of dependent random choice [13]. For $\omega \in \Omega$, put

$$d(\omega) = |\{i \in I : \omega \in E_i\}|.$$

Then $\mathbb{E}[d(\omega)/n] \geq \delta$. Let

$$G = \{\omega : d(\omega)/n \geq \delta/2\}.$$

Since $d/n \leq 1$,

$$\delta \leq \mathbb{E}[d/n] \leq \lambda(G) + (1 - \lambda(G))\delta/2,$$

so $\lambda(G) \geq \delta/2$.

Choose J uniformly from the s-subsets of I. For $\omega \in G$, $d(\omega) \geq \delta n/2$, and $s \leq \delta n/4$ gives

$$\mathbb{P}[J \subseteq \{i : \omega \in E_i\}] = \frac{\binom{d(\omega)}{s}}{\binom{n}{s}} \geq \left(\frac{\delta}{4}\right)^s.$$

Indeed each factor in $\prod_{r=0}^{s-1}(d(\omega) - r)/(n - r)$ is at least $(\delta n/2 - s)/n \geq \delta/4$. Averaging over ω yields

$$\mathbb{E}_J\lambda\Big(\bigcap_{j\in J} E_j\Big) \geq \lambda(G)\left(\frac{\delta}{4}\right)^s \geq \frac{\delta}{2}\left(\frac{\delta}{4}\right)^s.$$

Thus some J satisfies the claimed bound. The displayed lower bound has negative logarithm

$$\log(2/\delta) + s\log(4/\delta) = O_\delta(s),$$

which is the stated conditioning cost. $\square$

Lemma 10.2 (Residual repair pressure)**.** *Let $\mathcal{C}$ be a balanced two-collision code. Let ν be a law on distinct triples $(X, Y, Z) \in \mathcal{C}^3$ which are compatible on a retained block R, meaning that for every $r \in R$ the set $\{X_r, Y_r, Z_r\}$ has size 1 or 3. Put $L = [k] \setminus R$ and*

$$W_L(X, Y, Z) = \{t \in L : |\{X_t, Y_t, Z_t\}| = 2\}.$$

Then $|W_L(X, Y, Z)| \geq 1$ pointwise, and hence

$$\sum_{t \in L} \mathbb{P}_\nu[t \in W_L] \geq 1.$$

Proof. Because $\mathcal{C}$ is a two-collision code, every distinct triple of codewords has a coordinate with a $2 + 1$ split. Compatibility excludes such a split on R, so at least one split coordinate lies in L. Taking expectations gives the displayed inequality. $\square$

Lemma 10.3 (Public residual support promotion)**.** *In the setup of Lemma 10.2, let U be the current old support and put*

$$F = \{t \in L \setminus U : \mathbb{P}_\nu[t \in W_L] > 0\}.$$

The set F is determined by the current public law. Fix a constant $q \geq 1$. Then one of the following transitions is available:

(i) *fresh support promotion: $|F| > q$, in which case the public set F is promoted to old support and gives a potential drop $(B_f - B_o)|F|$;*

(ii) *bounded fresh residual: $|F| \leq q$, in which case exposing the full five-state equality pattern on F costs at most $q \log 5$. On each resulting atom, either the split part $F_{\mathrm{sp}} \subseteq F$ is nonempty, is promoted to old support, and produces a fresh close-pair block inside F_{sp}, or every residual split witness lies in the old support U.*

Proof. The probabilities $\mathbb{P}_\nu[t \in W_L]$ are functions of the current public law, so F is public. If $|F| > q$, adding all coordinates of F to the old support uses no private selector. Each such coordinate was fresh before the transition. In the potential of Section 16 this changes $\Phi_{\mathrm{fresh}} + \Phi_{\mathrm{old}}$ by $-(B_f - B_o)|F|$, which pays for the promotion and the bounded public bookkeeping attached to it.

Assume $|F| \leq q$. For every coordinate of F expose one of the five equality patterns

$$XYZ, \quad XY|Z, \quad XZ|Y, \quad YZ|X, \quad X|Y|Z.$$

There are at most $5^{|F|} \leq 5^q$ atoms. If some coordinates of F are split, let F_{sp} be the nonempty public split part on that atom. Promote F_{sp} to U before entering the close-pair child. Since $F_{\mathrm{sp}} \subseteq [k] \setminus U$, this gives a drop $(B_f - B_o)|F_{\mathrm{sp}}|$, which pays the bounded pattern exposure and deposits scale for the fresh close-pair block after the constant hierarchy is chosen. The constant $C_{\mathrm{fr}}(q)$ is chosen per promoted coordinate; since $|F_{\mathrm{sp}}| \geq 1$ and $|F| \leq q$, this single promoted coordinate already pays the whole bounded F-pattern transcript. One of the three split roles occupies at least one third of F_{sp}, giving a fresh close-pair block inside F_{sp} after relabelling. If no coordinate of F is split on the atom, then every residual split witness in L belongs to U, by the definition of F and the fact that W_L is nonempty pointwise under Lemma 10.2. This is the old-support branch. $\square$

Proposition 10.4 (Residual split-count DRC). *In the setup of Lemma 10.2, suppose*

$$\mathbb{E}_\nu|W_L| \geq \beta|L|$$

for $0 < \beta \leq 1$. *Then for every integer* $1 \leq s \leq \beta|L|/4$ *there is a block* $J \subseteq L$, $|J| = s$, *such that*

$$\nu[t \in W_L \textit{ for every } t \in J] \geq \frac{\beta}{2}\left(\frac{\beta}{4}\right)^s.$$

After exposing one role vector, there is a subblock $J' \subseteq J$ *with* $|J'| \geq |J|/3$ *and one ordered pair of samples, say* (X, Y) *after relabelling, such that on a subevent of mass at least*

$$\frac{\beta}{2}\left(\frac{\beta}{12}\right)^s$$

one has

$$X_t = Y_t \neq Z_t \qquad (t \in J').$$

Proof. Apply Lemma 10.1 to the probability space of triples under ν, the index set L, and the events $E_t = \{t \in W_L\}$. This gives the block J and the first mass bound.

On the event that every $t \in J$ splits, each coordinate has one of three roles: $XY|Z$, $XZ|Y$, or $YZ|X$. There are at most 3^s role vectors, so some role vector has conditional mass at least 3^{-s}. On this atom, one role appears on at least $s/3$ coordinates; call that majority subblock J'. Relabeling the samples gives the displayed close-pair relation. □

Proposition 10.5 (Residual decision order). *At a residual node, after exact split exposure on an active block* I, *put* $R = I \setminus D$, $L = [k] \setminus R$, *and*

$$S = \mathbb{E}_\nu|W_L|.$$

Fix constants $\beta > 0$ *and* $q \geq 1$. *The residual transition is run in the following public order:*

(i) *apply Lemma 10.3 to the fresh residual support;*

(ii) *if that lemma produces a bounded fresh close-pair block, leave residual mode through close-pair mode;*

(iii) *otherwise the branch is old-supported, meaning every residual split witness is contained in the current old support* U *pointwise; no random old positive-support selector is exposed;*

(iv) *on an old-supported branch, if* $S \geq \beta|L|$ *and* $|L| \geq 4/\beta$, *apply Proposition 10.4 with*

$$s_L = \max\{1, \lfloor \beta|L|/8 \rfloor\};$$

the resulting role block is old and is registered in the old-window ledger;

(v) *on an old-supported branch with* $S \geq \beta|L|$ *but* $|L| < 4/\beta$, *expose the bounded set* $W_L \subseteq L$ *directly and route the bounded support through the old-incidence or bounded-leaf ledger;*

(vi) *on an old-supported branch with* $S < \beta|L|$, *apply Lemma 13.5 to the deterministic old positive support. On each old-pattern atom the resulting split set* K *is classified by Proposition 10.8: deletion-trapped atoms give scale descent, while the remaining old atoms enter the old-window/close-pair ledger.*

No DRC block is used unless the linear split-count hypothesis $S \geq \beta|L|$ is satisfied and the resulting DRC range contains a nonempty block.

Proof. The fresh residual support F and the number S are functions of the current public law, so the case distinction is public. If $|F| > q$, Lemma 10.3 promotes every coordinate of F to the global old support. After this update, every coordinate with positive probability of belonging to W_L is old. If $|F| \leq q$, the same lemma exposes only the five-state pattern on F, at cost at most $q \log 5$; each atom either promotes its nonempty split part and gives a fresh close-pair block paid by that bounded fresh drop, or leaves all residual witnesses in U.

Thus no residual branch exposes the whole random set $W_L \subseteq L$ as an unpaid sparse selector. On the remaining old-supported branches, the dense case $S \geq \beta|L|$ with $|L| \geq 4/\beta$ satisfies the non-vacuous hypothesis of Proposition 10.4 for the displayed value of s_L. Indeed $s_L \geq 1$ and $s_L \leq \beta|L|/4$. Since every split witness is old, the DRC block and its majority role block are old blocks and are entered into the old-window incidence ledger.

If $S \geq \beta|L|$ but $|L| < 4/\beta$, then the whole residual domain has bounded size. Exposing the exact subset $W_L \subseteq L$ and its role pattern has bounded entropy depending only on β. Because the branch is old-supported, this bounded support is registered in the old ledger or stops through the bounded-leaf/high-revisit exit. No DRC block is needed in this bounded case.

If $S < \beta|L|$, the branch is still old-supported and Lemma 13.5 applies directly to the deterministic old positive support $O = \{u \in U : \mathbb{P}[u \in W_L] > 0\}$, which is a function of the public law rather than a random selector. This exposes only old-coordinate pattern data and charges it to old incidences. On each resulting atom the old split set $K = W_L$ is known inside the already paid old pattern. Proposition 10.8 then separates deletion-trapped scale descents from old close-pair returns. The last sentence is part of the stated order and prevents the non-linear residual pressure $S \geq 1$ from being used as a linear DRC input. □

Lemma 10.6 (Fresh-old-deletion trichotomy)**.** *Let J be the current block, $D \subseteq J$ the deletion set, U the old support, and let*

$$K \subseteq D \cup ([k] \setminus J)$$

have size t. Define

$$\mathrm{Fresh}(K) = K \setminus (J \cup U), \qquad \mathrm{Old}(K) = K \cap U.$$

Then at least one of the following holds:

$$t \leq 4|D|, \qquad |\mathrm{Fresh}(K)| \geq t/4, \qquad |\mathrm{Old}(K)| \geq t/2.$$

Proof. If $t \leq 4|D|$, the first alternative holds. Hence assume that $t > 4|D|$. Since $K \cap J \subseteq D$, one has $|K \setminus J| > 3t/4$. If also $|\mathrm{Fresh}(K)| < t/4$, then the remaining coordinates of $K \setminus J$ lie in U, giving

$$|\mathrm{Old}(K)| \geq |K \setminus J| - |\mathrm{Fresh}(K)| > t/2.$$

No disjointness assumption between U and J is used. The partition is only on $K \setminus J$: on this set $(K \setminus J) \setminus U = \mathrm{Fresh}(K)$, while $(K \setminus J) \cap U \subseteq \mathrm{Old}(K)$. □

Proposition 10.7 (Explicit residual split atom transition)**.** *At a residual node, let I be the current active block and let $D = \mathrm{Split}_I(X, Y, Z)$ be the exact split set already exposed on the branch. Put $R = I \setminus D$ and $L = [k] \setminus R$. Suppose that a further public atom exposes*

$$K = W_L(X, Y, Z) \subseteq L, \qquad |K| = t.$$

Then $K \subseteq D \cup ([k] \setminus I)$ and every coordinate of K is split. With respect to the current old support U, at least one of the following transitions is available:

(1) *deletion-trapped:*

$$t \le 4|D|;$$

(2) *fresh escape:*

$$|\operatorname{Fresh}(K)| \ge t/4;$$

(3) *old close-pair return: after routing $K \cap U$ through Lemma 13.5, there is $O' \subseteq K \cap U$ with $|O'| \ge t/6$ and one ordered pair of samples, say (X, Y) after relabelling, such that*

$$X_{O'} = Y_{O'}, \qquad Z_i \ne X_i \quad (i \in O').$$

Proof. Since $L = [k] \setminus (I \setminus D) = D \cup ([k] \setminus I)$, the exposed atom has $K \subseteq D \cup ([k] \setminus I)$. Apply Lemma 10.6 with the current block I and deletion set D. The first two alternatives of that lemma give the deletion-trapped and fresh escape branches.

In the old branch, $|K \cap U| \ge t/2$. Each coordinate of $K \cap U$ is in W_L, hence is split. Lemma 13.5 applies to the public old block $K \cap U$ and gives a close-pair subblock of size at least $(t/2)/3 = t/6$, with its role cost entered into the old-window ledger. □

Proposition 10.8 (Sparse-exposed residual transition)**.** *Let a law on distinct triples (X, Y, Z) satisfy*

$$|\operatorname{Split}_I(X, Y, Z)| \le s$$

on a positive-mass branch. Expose the exact split set $D = \operatorname{Split}_I(X, Y, Z)$, so $|D| \le s$ and the triples are compatible on $I \setminus D$. Suppose residual repair produces a split block

$$K \subseteq D \cup ([k] \setminus I), \qquad |K| = t.$$

With respect to an old support U, at least one of the following holds:

(1) *deletion-trapped:*

$$t \le 4s;$$

(2) *fresh escape:*

$$|\operatorname{Fresh}(K)| \ge t/4;$$

(3) *old close-pair return: after routing $K \cap U$ through Lemma 13.5, there is $O' \subseteq K \cap U$ with $|O'| \ge t/6$ and one ordered pair of the samples, say (X, Y) after relabelling, such that*

$$X_{O'} = Y_{O'}, \qquad Z_i \ne X_i \quad (i \in O').$$

Proof. Apply Lemma 10.6 with the deletion set D. Its first branch gives $t \le 4|D| \le 4s$, and its second branch is the fresh escape branch.

In the third branch, $|K \cap U| \ge t/2$. Since K is a split block, every coordinate in $K \cap U$ is split. Lemma 13.5 applies to the public old block $K \cap U$ and gives a close-pair subblock of size at least $(t/2)/3 = t/6$. □

Lemma 10.9 (Deletion-trapped scale descent)**.** *In the setup of Proposition 10.7 or Proposition 10.8, assume $|I| = m$, $|D| \le s = \sigma m$, and the deletion-trapped old-only branch occurs. Then every close-pair block produced inside the residual block has size at most*

$$4\sigma m.$$

In particular, if $4\sigma < 1$, repeated pure deletion-trapped old-only returns strictly decrease scale geometrically.

Proof. The deletion-trapped branch gives $|K| = t \le 4|D| \le 4s = 4\sigma m$. Any close-pair subblock selected inside K has size at most $|K|$. Iterating the same estimate gives geometric scale decay when $4\sigma < 1$. $\square$

Lemma 10.10 (Nontrapped residual blocks escape the current block)**.** *In the setup of Proposition 10.7 or Proposition 10.8, if the residual block is not deletion-trapped, so $t > 4|D|$, then*

$$|K \setminus I| > 3t/4.$$

If all coordinates of K are old and split, then after role refinement there is an old close-pair block $L \subseteq K \setminus I$ with

$$|L| > t/4.$$

Proof. Since $K \subseteq D \cup ([k] \setminus I)$, every coordinate of K lying in I lies in D. Hence $|K \cap I| \le |D| < t/4$, and

$$|K \setminus I| > 3t/4.$$

On $K \setminus I$, one of the three split roles occurs on more than $t/4$ coordinates. That majority-role set is the required close-pair block. $\square$

11 Public DRC Blocks and Residual Repair

DRC-generated blocks must be chosen publicly.

Definition 11.1 (Public DRC convention)**.** *Whenever a DRC argument proves the existence of a coordinate block with a given mass lower bound, the proof chooses the first valid block in a fixed public order. Thus the block is a deterministic function of the current public law and the fixed parameters.*

Under this convention residual repair blocks are public. The only private data left in old-coordinate residuals is the role vector

$$\mathrm{Role}_O \in \{XY|Z, XZ|Y, YZ|X\}^O.$$

Exposing this vector costs at most $|O| \log 3$, and a majority role gives a close-pair subblock O' with $|O'| \ge |O|/3$. Hence the role exposure is locally amortized by the block it creates.

Proposition 11.2 (Residual old blocks are public up to role entropy)**.** *In the residual repair route of Proposition 10.4, enforce the public DRC convention. Then every old close-pair parent block has the following form:*

(i) *a residual DRC block K is deterministic from the current public law;*

(ii) *its old part $O = K \cap U$ is public;*

(iii) *the only remaining private selector is a role vector on O, whose entropy is at most $|O| \log 3$;*

(iv) *after role refinement, the produced close-pair block has size at least $|O|/3$.*

Proof. The DRC block is public by the deterministic tie-break. Since I, U, K are public at that point, the set $O = K \cap U$ is public as well. The role vector has at most $3^{|O|}$ values. Conditioning on one role vector costs at most $|O| \log 3$, and a majority role occupies at least one third of O. $\square$

12 Close-Pair Product Lemmas

Lemma 12.1 (Conditioned product factorization)**.** *For each $j \in J$, let p_j, q_j be probability laws on Ω_j, and put*

$$\gamma_j = \sum_a p_j(a) q_j(a) < 1.$$

Let $Q = \bigotimes_{j \in J}(p_j \otimes q_j)$ on pairs (A_J, B_J) and let

$$E = \{A_j \neq B_j \text{ for all } j \in J\}.$$

Then $Q(\cdot \mid E)$ factorizes as

$$Q_E = \bigotimes_{j \in J} Q_j^*, \qquad Q_j^*(a, b) = \frac{p_j(a) q_j(b) \mathbf{1}_{a \neq b}}{1 - \gamma_j}.$$

Proof. The event E is the intersection of coordinate events $E_j = \{A_j \neq B_j\}$, and under Q these coordinate pairs are independent. Thus $Q(E) = \prod_j (1 - \gamma_j)$, and division by this product gives the displayed coordinate factorization. □

Lemma 12.2 (Product transfer to near-sunflowers)**.** *Let $P = \mathrm{Law}(A_J, B_J)$ be supported on $A_j \neq B_j$ for every $j \in J$. Let $p_j = \mathrm{Law}(A_j)$, $q_j = \mathrm{Law}(B_j)$, and let Q_E be the conditioned product law of Lemma 12.1. Assume*

$$D(P \| Q_E) \leq \epsilon, \qquad \gamma_j \leq \gamma_0 < 1 \quad (j \in J),$$

and

$$\frac{1}{|J|} \sum_{j \in J} \mathrm{Coll}(p_j) \leq \alpha.$$

If $A^{(1)}, A^{(2)}, A^{(3)}$ are independent samples from the true A-marginal P_A, then

$$\mathbb{P}\left[|\operatorname{Split}_J(A^{(1)}, A^{(2)}, A^{(3)})| \leq \frac{6\alpha |J|}{(1 - \gamma_0)^2} \right] \geq \frac{1}{2} - 3\sqrt{\epsilon/2}.$$

Proof. Under Q_E, the A_j-marginal is

$$p_j^*(a) = \frac{p_j(a)(1 - q_j(a))}{1 - \gamma_j}.$$

Since $1 - q_j(a) \leq 1$ and $1 - \gamma_j \geq 1 - \gamma_0$,

$$\mathrm{Coll}(p_j^*) = \sum_a \frac{p_j(a)^2 (1 - q_j(a))^2}{(1 - \gamma_j)^2} \leq \frac{\mathrm{Coll}(p_j)}{(1 - \gamma_0)^2}.$$

Thus three independent samples from the A-marginal of Q_E have expected split count at most

$$\frac{3\alpha |J|}{(1 - \gamma_0)^2}$$

by Lemma 6.2. Markov's inequality gives probability at least $1/2$ for the displayed event under that product marginal.

By Pinsker's inequality [6,8,18],

$$\mathrm{TV}(P, Q_E) \le \sqrt{\epsilon/2}.$$

Total variation cannot increase under marginalization, and the distance between threefold product laws is at most three times the one-sample distance. Hence the same event has probability at least $1/2 - 3\sqrt{\epsilon/2}$ under P_A^3. $\square$

Lemma 12.3 (Projection near-sunflowers lift to code triples)**.** *Let μ be a law on a code $\mathcal{C}$, let π_J be projection to J, and let $\nu = (\pi_J)_*\mu$. If a projected event $F \subseteq (\pi_J\mathcal{C})^3$ has*

$$\nu^3(F) \ge \rho,$$

then at least one of the following holds:

(i) *the first codeword x in the fixed public order satisfying $\mu(x) \ge \rho/6$ is retained as a terminal codeword atom;*

(ii) *for independent $X, Y, Z \sim \mu$, the event $(\pi_J X, \pi_J Y, \pi_J Z) \in F$ occurs with X, Y, Z distinct with probability at least $\rho/2$.*

Proof. The projections of independent samples from μ are independent samples from ν, so the projected event has probability exactly ρ at the code level before imposing distinctness. Let

$$\Delta = \mathbb{P}[X = Y] = \sum_x \mu(x)^2.$$

The probability that X, Y, Z are not all distinct is at most 3Δ. If $\Delta \ge \rho/6$, then $\max_x \mu(x) \ge \Delta \ge \rho/6$. The retained atom is chosen by the fixed public order and the discarded complement is charged by the terminal-pruning convention, so no nonterminal child inherits that complement. Otherwise $3\Delta < \rho/2$, and the distinct lift has mass at least $\rho/2$. $\square$

Theorem 12.4 (Close-pair master exit)**.** *Let μ be the current one-sample marginal law on $\mathcal{C}$ at a tuple node, and let J be a public close-pair block. Suppose $P = \mathrm{Law}(A_J, B_J)$ is supported on*

$$A_j \ne B_j \qquad (j \in J),$$

where the A-marginal is the projection of the current marginal law: $A_J = \pi_J(X)$ for $X \sim \mu$. Let

$$p_j = \mathrm{Law}(A_j), \qquad q_j = \mathrm{Law}(B_j), \qquad \gamma_j = \sum_a p_j(a) q_j(a),$$

and, if all $\gamma_j < 1$, let Q_E be the conditioned product law from Lemma 12.1.

Fix parameters $h, \alpha, \epsilon, \gamma_0$ with $0 \le \alpha \le 1$, $0 \le \gamma_0 < 1$, and

$$\rho = \frac{1}{2} - 3\sqrt{\epsilon/2} > 0.$$

Then at least one of the following holds:

(1) *low common-symbol entropy:*

$$\mathbf{H}(A_J) \le h|J|;$$

(2) *cross-collision:*

$$\gamma_j > \gamma_0 \quad \textit{for some } j \in J;$$

(3) *product KL excess (only in the case all $\gamma_j < 1$):*

$$D(P\|Q_E) > \epsilon;$$

(4) *high A-side collision:*

$$\frac{1}{|J|}\sum_{j\in J}\mathrm{Coll}(p_j) > \alpha;$$

(5) *terminal codeword atom: the first codeword in the fixed public order with*

$$\mu(x) \ge \rho/6$$

is retained as a terminal atom;

(6) *law-level near-sunflower return: for independent $X, Y, Z \sim \mu$,*

$$\mathbb{P}\left[X, Y, Z \text{ distinct and } |\operatorname{Split}_J(X,Y,Z)| \le \frac{6\alpha|J|}{(1-\gamma_0)^2}\right] \ge \rho/2.$$

Proof. If some $\gamma_j = 1$, then the cross-collision alternative holds because $\gamma_0 < 1$. Hence we may assume all $\gamma_j < 1$, so Q_E is defined by Lemma 12.1.

If any of the first four alternatives holds, there is nothing to prove. Assume they all fail. Then

$$D(P\|Q_E) \le \epsilon, \qquad \gamma_j \le \gamma_0 \quad (j \in J),$$

and

$$\frac{1}{|J|}\sum_{j\in J}\mathrm{Coll}(p_j) \le \alpha.$$

Lemma 12.2 gives projected near-sunflower mass at least ρ for three independent samples from the A-marginal. Since this A-marginal is the J-projection of μ, Lemma 12.3 gives either a terminal codeword atom of mass at least $\rho/6$ or a distinct law-level lift of mass at least $\rho/2$. These are alternatives (5) and (6). In the tuple proof tree, the latter is realized by conditioned copying from the same public key, not by asserting an ambient-code density increment. $\square$

Theorem 12.5 (Positive KL public stopping scan)**.** *Let*

$$V = (V_1, \dots, V_m)$$

be a list of tuple-coordinate variables, ordered by a public order; each V_t is one actual sample coordinate value, with its sample label and ambient coordinate determined by the public list. Let $P \ll Q$ be laws on the product space of V with $D(P\|Q) > 0$. Fix a prefix cost-density threshold $\lambda > 0$, a collision threshold $0 < \tau < 1$, and a tail threshold $0 < \rho \le 1/2$.

Then the following public stopping scan gives an exhaustive list of retained or charged-pruned children at the current tuple node. Along a realized branch, let t be the first index, if one exists, whose current conditional one-coordinate KL is positive, and let i be the first index, if any, for which the realized prefix $V_{\le i} = u_{\le i}$ has

$$-\log P(V_{\le i} = u_{\le i}) > \lambda i.$$

If the positive-KL index t occurs before such a crossing, the branch is routed as in (1) below. If a first crossing occurs before the positive-KL stop, the branch is routed as in (2). If neither stop occurs through the end of the list, the full tuple prefix is terminalized as a projected shadow with

$$-\log P(V = v) \le \lambda m.$$

In the first two cases the output is one of the following:

(1) *an amortized projected prefix $V_{<t} = u$ with*

$$-\log P(V_{<t} = u) \le \lambda(t-1),$$

followed by a one-coordinate positive-KL tail. After conditioning on the public positive tail of mass at least ρ, or after pruning a positive tail of mass $< \rho$ at cost at most 2ρ, this gives either a terminal projected tuple pin of conditional mass at least $\rho\tau$ in the sense of Lemma 8.2 or a low-collision conditional hash law whose paid prefix and tail/pruning transcript is within the fixed hash-window budget. If the set-valued tail/pruning increment would overflow the budget, the branch stops on the already paid value prefix instead of treating that set event as a projected shadow;

(2) *an earlier first-crossing coordinate i with an amortized prefix $V_{<i} = u_{<i}$. At this coordinate the proof peels the public heavy atoms of the conditional law. Heavy-atom branches terminalize as projected tuple pins of conditional mass at least $\rho\tau$; the light complement is either a charged pruning branch of mass $< \rho$, or, after conditioning on that complement, a low-collision conditional hash law whose paid prefix and complement transcript is within the fixed hash-window budget. If the complement event would overflow the budget, the branch stops on the already paid value prefix.*

All coordinate identities and sample labels used here are determined by the public order and the charged prefix value. The conclusion is tuple-level: a pin is terminalized as a projected shadow including the amortized prefix used to reach it, with mixed-label prefixes projected by Lemma 8.3, and a hash law remains inside the paid public transcript and the hash window of that sample label until conditioned copying is invoked. Set-valued tail and pruning events are retained only inside this paid hash transcript; terminal projected shadows contain only coordinate-value pins. Every collision statement in the conclusion is for the actual P-conditional law of the selected tuple-coordinate variable under the paid prefix and tail/pruning event; the reference law Q is used only to locate a positive-KL tail. First crossings and the final full-prefix terminal branch are driven by the P-prefix mass, not by any low-collision claim about Q. If an amortized prefix is nonempty but exceeds the fixed hash-window transcript budget, the branch terminalizes on that projected tuple-coordinate value prefix instead of entering hash mode.

Proof. For a realized branch write

$$C_r = -\log P(V_{\le r} = v_{\le r}).$$

By Lemma 6.5, the conditional KL functions

$$d_t(u) = D(P(V_t \in \cdot \mid V_{<t} = u) \| Q(V_t \in \cdot \mid V_{<t} = u))$$

are public after the prefix value u is paid. Scan the realized prefixes in the fixed public order. If the first positive-KL index t occurs before any prefix crossing, then $C_{t-1} \le \lambda(t-1)$ and the prefix $u = V_{<t}$ is amortized. Under that prefix, the one-coordinate laws p, q satisfy

$$D(p\|q) > 0.$$

The positive tail $S_+ = \{a : p(a) > q(a)\}$ has positive p-mass by Lemma 6.8. If $p(S_+) \ge \rho$, Lemma 6.6 applied to S_+ gives either a value atom of conditional mass at least $\rho\tau$ or a low-collision conditional law. The value atom is terminalized by Lemma 8.2. A low-collision law enters hash mode only after the public tail event $V_t \in S_+$ is entered into the transcript; the collision bound is for $P(V_t \in \cdot \mid V_{<t} = u,\ V_t \in S_+)$, not for the corresponding Q-conditional law. This costs at most $\log(1/\rho)$ and is a fixed part of the window budget. If the already paid prefix together with this fixed tail cost exceeds the fixed window budget, the set event $V_t \in S_+$ is not appended; the branch terminalizes on the projected value prefix $V_{<t} = u$ instead. The constant hierarchy chooses B_{win} larger than every fixed tail/pruning increment, so an empty prefix never causes this overflow. If $p(S_+) < \rho$, prune S_+ at cost at most 2ρ by Lemma 6.8. This means that the branch $V_t \in S_+$ is an explicit pruning branch with charged discarded mass, while the retained branch is the public complement $V_t \notin S_+$. The remaining public set has p-mass at least $1 - \rho \geq \rho$, so applying Lemma 6.6 there gives the same value-pin/hash alternative, now under the public pruned event $V_t \notin S_+$ and again for the actual P-conditional law, provided the fixed pruning increment fits in the remaining window budget. If it does not fit, the branch terminalizes on the projected value prefix $V_{<t} = u$ before the set event is appended. This is alternative (1).

If a prefix crossing occurs before the positive-KL stop, or if no positive-KL stop has occurred before the first crossing, let i be the first index with $C_i > \lambda i$. Then the previous prefix is amortized: $C_{i-1} \leq \lambda(i-1)$. Let p be the one-step conditional law of V_i under that previous prefix; this law is attached to the public coordinate position i. Apply Lemma 6.7 to this law. The heavy set

$$H = \{a : p(a) \geq \rho\tau\}$$

has bounded size, and each branch $V_i = a \in H$ is a genuine terminal value pin of conditional mass at least $\rho\tau$. These branches are converted to projected tuple pins by Lemmas 8.2 and 8.3 together with the amortized prefix. On the light complement L, if $p(L) < \rho$ then the branch $V_i \in L$ is an explicit charged pruning branch and is not inherited by a cylinder child. If $p(L) \geq \rho$, then the conditional law of V_i on L has collision at most τ. This gives a low-collision hash coordinate carried by the paid prefix and the public complement event $V_i \in L$, provided that transcript remains inside the fixed window budget. If it would overflow the budget, the set event is not appended and the branch terminalizes on the already paid value prefix $V_{<i} = u_{<i}$. This is alternative (2).

It remains to cover branches on which neither a positive-KL stop nor a prefix crossing occurs. Then $C_m \leq \lambda m$, so the full tuple-coordinate value $V = v$ is an amortized projected shadow. This terminal branch is included in the shadow ledger and has no hash output. Thus the scan is exhaustive: every realized branch either stops by positive KL, by first crossing, by an explicit charged pruning branch inside those cases, or by the final amortized full prefix. $\square$

Proposition 12.6 (Product-KL excess has a one-coordinate public exit). *In every product-KL excess branch in Theorem 12.4, the excess*

$$D(P \| Q_E) > \epsilon$$

can be routed, using only the current public tuple state and paid transcript, to one of:

(i) *a terminal projected tuple shadow whose local entropy cost is charged by the proof-tree ledger;*

(ii) *a public low-collision hash coordinate for an actual sample coordinate, not merely a low-collision law on the pair alphabet (A_j, B_j).*

Proof. List the public close-pair block as $J = (j_1, \ldots, j_m)$. In close-pair mode, after relabelling the active samples, the tuple coordinates on J have the form

$$X_J = Y_J = A_J, \qquad Z_j = B_j \neq A_j \quad (j \in J).$$

Form the publicly ordered interleaved tuple-coordinate list

$$V = (A_{j_1}, B_{j_1}, A_{j_2}, B_{j_2}, \ldots, A_{j_m}, B_{j_m}).$$

This is only a bijective reordering of the pair data (A_J, B_J). Therefore, if P_V, Q_V are the pushforwards of P, Q_E under this interleaving, then

$$D(P_V \| Q_V) = D(P \| Q_E) > \epsilon > 0.$$

Apply Theorem 12.5 to the laws P_V, Q_V and to this public list. Its stopping scan is exhaustive on the realized tuple-prefix tree. On every hash alternative, the selected local variable is now an actual tuple-coordinate value: either the common value $A_{j_r} = X_{j_r} = Y_{j_r}$, or the third-sample value $B_{j_r} = Z_{j_r}$. The prefix variables are also actual tuple-coordinate values in the same public list. The low-collision alternative produced by Theorem 12.5 is a statement about the actual P_V-conditional law of this selected tuple coordinate under the paid prefix and tail/pruning transcript. The reference product law Q_V is not passed to the hash ledger.

If Theorem 12.5 terminalizes an amortized prefix, the final amortized full tuple prefix, a positive-tail value atom, or a first-crossing heavy atom, the corresponding sample-coordinate values are entered into the terminal projected shadow and paid by the proof-tree entropy ledger; mixed-label prefixes are grouped into multi-sample shadows by Lemma 8.3. First-crossing light complements are either charged pruning branches or paid set events inside the hash transcript; they are not terminal value-shadows. If the theorem produces a low-collision conditional law, that law is attached to the public sample-coordinate label (X, j_r) (equivalently (Y, j_r)) or (Z, j_r) under the paid public conditioning transcript. The branch enters the hash window for that public sample label only, and only while the paid transcript is inside the fixed hash-window budget.

Thus the product-KL branch is routed through the KL chain rule on the actual tuple-coordinate list. The proof does not choose an arbitrary first coordinate of J, and it does not use a low-collision statement on the pair alphabet. □

13 Old Support and Projection

The old support cannot be paid as an $\exp(O(|U|))$ overhead at every rank. That would iterate to $\exp(\Theta(k^2))$. The correct statement is a projection-or-terminalization criterion.

Definition 13.1 (Old projection). *Old role/support analysis data satisfies projection if every bit of that data is one of the following:*

(i) *terminal-local, and used only to define a terminal block of comparable size;*

(ii) *public-progress, affecting only rank/fresh/scale/high-revisit progress;*

(iii) *not retained before terminal shadow counting;*

(iv) *promoted to public state with entropy charged to a telescoping potential.*

Lemma 13.2 (Old projection implies shadow control). *If every old-window segment satisfies old projection, then the old component of the terminal shadow has entropy*

$$\mathbf{H}(\Theta_{\text{old}}) \leq C\, \mathbb{E}[T_{\text{old}}(\Theta) + R_{\text{old}}(\Theta)].$$

Proof. Only terminal-local old data can influence Θ_{old} after projection. For one terminal old block, local role exposure costs $O(|O'|)$, and the value selector is allowed only in the low-symbol case $\mathbf{H}(A \mid E) \leq h|O'|$ of Theorem 13.6. Hence the terminal value contribution is $O(|O'|)$. High-symbol old value vectors are not exposed; they are routed as nonterminal old-window data and are not retained in the terminal shadow unless a later low-symbol terminalization occurs. Summing over terminal old exits gives the bound. Public-progress and unretained data do not contribute to Θ_{old}. □

Lemma 13.3 (High revisit local pressure)**.** *At a close-pair node, suppose a coordinate i has role*

$$X_i = Y_i \neq Z_i$$

and let $A_i = X_i = Y_i$ under the current conditional law. For every $0 < \rho \leq 1/2$ and $0 < \tau < 1$, the following exhaustive public routing is available. Put

$$H = \{a : \mathbb{P}[A_i = a] \geq \rho\tau\}, \qquad L = \Omega_i \setminus H.$$

Then $|H| \leq (\rho\tau)^{-1}$, every branch $A_i = a \in H$ is a terminal one-coordinate projected tuple pin of conditional mass at least $\rho\tau$, and the complement satisfies exactly one of:

(i) $\mathbb{P}[A_i \in L] < \rho$, *in which case the complement is a charged pruning branch of loss at most 2ρ;*

(ii) $\mathbb{P}[A_i \in L] \geq \rho$, *in which case, after conditioning on the public set event $A_i \in L$, the conditional law has collision at most τ and therefore gives a public low-collision hash coordinate.*

Proof. This is Lemma 6.7 applied to the one-coordinate law of A_i. The only point to note is that the coordinate identity and the sets H, L are determined by the current public conditional law. If L is retained, the set event $A_i \in L$ is appended only as part of the paid hash transcript; if it would exceed the remaining hash-window budget, the branch terminalizes before entering the hash window, as in Theorem 12.5. □

Lemma 13.4 (Old-pattern container)**.** *Fix an old support U and an ordered triple (X, Y, Z). For each $u \in U$, record the equality partition of $\{X_u, Y_u, Z_u\}$; there are five possibilities:*

$$XYZ, \quad XY|Z, \quad XZ|Y, \quad YZ|X, \quad X|Y|Z.$$

After conditioning on one atom of this five-state pattern on U, every future old split block $O \subseteq U$ has a deterministic subblock $O' \subseteq O$ with $|O'| \geq |O|/3$ and one ordered pair, say (X, Y) after relabelling, such that

$$X_{O'} = Y_{O'}, \qquad Z_o \neq X_o \quad (o \in O').$$

Proof. On a fixed five-state atom, each coordinate of a split block has one of the three roles $XY|Z, XZ|Y, YZ|X$. One role occurs on at least one third of the coordinates. The corresponding majority set is O'. □

Lemma 13.5 (Old-supported residual witness routing)**.** *At an old-supported residual node, suppose $W_L \subseteq U$ pointwise and W_L is nonempty pointwise. Let*

$$O = \{u \in U : \mathbb{P}_\nu[u \in W_L] > 0\}.$$

Then O is public. Before any nonterminal continuation, O is registered in the old-window incidence ledger and the five-state equality pattern on O is exposed. On every resulting positive-mass atom, *either the registration triggers the high-revisit exit, or the atom determines the old split set $K = W_L \subseteq O$, which is nonempty, and produces an old close-pair block $K' \subseteq K$ with*

$$|K'| \geq |K|/3$$

after relabelling. The pattern cost is charged to the registered old incidences and hence decreases Φ_{old} in the non-high-revisit case.

The same conclusion holds if a public old block $K \subseteq U$ is already known to be split on the branch; then one registers only K and exposes its three-role vector.

Proof. The set O is determined by the current public law. Register the incidences of O and expose the five-state pattern

$$XYZ, \quad XY|Z, \quad XZ|Y, \quad YZ|X, \quad X|Y|Z$$

on O. Since $W_L \subseteq O$ and W_L is nonempty pointwise, the split coordinates on any positive-mass atom are exactly a nonempty set $K = W_L \subseteq O$. If a coordinate crosses the revisit threshold, the exposed split label gives a close-pair revisit of that coordinate; the high-revisit exit is then the exhaustive terminal/pruning/hash routing of Lemma 13.3. Otherwise these incidences are debited in N_{old}, so they lower Φ_{old} in Section 16. Among the split coordinates in K, one of the three roles $XY|Z, XZ|Y, YZ|X$ occurs on at least one third; relabelling the samples gives the close-pair block K'.

If a public split block K is already available, the same proof skips the five-state exposure on O and exposes only the three-role vector on K. □

Theorem 13.6 (Old-symbol values are terminal-local)**.** *Fix an old support U and condition on one old-pattern atom. In the public normal form, common old-symbol values are exposed only on terminal low-symbol branches. More precisely, if a node with event E obtains, after relabelling, an old close-pair block O' and*

$$A = X_{O'} = Y_{O'},$$

then exactly one of the following happens.

(i) *The branch is low-symbol:*

$$\mathbf{H}(A \mid E) \leq h\,|O'|.$$

The value selector A may be exposed, and the branch immediately terminates as a projected shadow which pins the coordinates O'.

(ii) *The branch is high-symbol. The proof records only role, equality-pattern, scale, and block-incidence data; the value vector A is not exposed and is not part of the terminal transcript. The branch is routed to the fixed-scale old-window analysis of Theorem 14.3. If a separate close-pair exit, such as Proposition 12.6, exposes a bounded tuple-coordinate prefix before entering hash mode, that prefix is charged to the shadow/hash ledger of that exit and not to the old-window ledger.*

Consequently the old-symbol value contribution to the terminal proof-tree entropy is bounded by

$$\sum_{\text{terminal low-symbol } v} \mathbb{P}(E_v)\,\mathbf{H}(A_v \mid E_v) \leq h\,\mathbb{E}T_{\mathrm{old}}(\Theta),$$

where T_{old} is the number of old coordinates pinned by terminal old-symbol shadows. No alphabet-dependent term such as $\mathbf{H}(X_U, Y_U, Z_U)$ is used in the global charge.

Proof. The theorem is a normal-form rule plus its entropy accounting. On a low-symbol branch the child selector is the terminal value vector A_v (and the finite role/block data already charged by Lemma 13.4 or Proposition 13.7). Its conditional entropy is at most $h|O'_v|$, and the branch stops with O'_v included in the projected shadow. Summing over terminal leaves gives the displayed bound because $T_{\mathrm{old}}(\Theta)$ counts these pinned coordinates with multiplicity in the same terminal shadow ledger.

On a high-symbol old-window continuation the vector A_v is not a selector of the old-window proof tree. Only finite role, pattern, scale, and incidence information is revealed before the branch is passed to Theorem 14.3. Those non-value transcripts are paid by the old-incidence and scale ledgers. If another exit later exposes actual coordinate values, it does so through its own terminal-shadow or bounded hash-prefix rule; such values are not charged to the old ledger. Therefore no alphabet-dependent old-symbol value entropy remains in the old-window account. $\square$

Proposition 13.7 (Old close-pair returns are public up to paid old data)**.** *Suppose a residual transition produces an old block $O \subseteq U$ after an exact split-set atom and a public DRC choice have been fixed. Then the resulting old close-pair return either:*

(i) *is obtained from a previously fixed old-pattern atom by Lemma 13.4, so no new coordinate selector is private;*

(ii) *or pays local role entropy $O(|O'|)$ to produce a close-pair block $O' \subseteq O$ with $|O'| \geq |O|/3$.*

In either case the block can be assigned to its public dyadic scale and fed to the fixed-scale old-window analysis.

Proof. The exact split-set atom and the public DRC convention make the parent block O a function of the paid public state. If an old-pattern atom on U is already fixed, Lemma 13.4 chooses the majority-role subblock deterministically from that atom. Otherwise expose the local role vector on O; it has at most $3^{|O|}$ values, and a majority role gives $|O'| \geq |O|/3$, so the exposure cost is $O(|O'|)$. The resulting coordinate set has a definite size and is placed in the corresponding fixed-scale old window. $\square$

14 Fixed-Scale Old Windows

The last old-support mechanism is a fixed-scale high-symbol old-role window.

Lemma 14.1 (Pooling with a fixed bounded reservoir)**.** *Fix a reservoir $W = (X^1, \ldots, X^r)$ throughout a window. Let $B_1, \ldots, B_R$ be split blocks and suppose each coordinate appears in at most Q of the blocks. Put $N = \sum_b |B_b|$. For every incidence (b, i) with $i \in B_b$, suppose the branch determines a split label $(s, t|u)$, meaning*

$$X^s_i = X^t_i \neq X^u_i.$$

Then there is one label $(s, t|u)$ and a coordinate set L such that

$$X^s_L = X^t_L, \qquad X^u_\ell \neq X^s_\ell \quad (\ell \in L),$$

and

$$|L| \geq \frac{N}{M_r Q}, \qquad M_r = 3\binom{r}{3}.$$

Proof. There are N incidences and at most M_r possible split labels. Some label occurs at least N/M_r times. Since each coordinate appears in at most Q blocks, the set of coordinates carrying this fixed label has size at least $N/(M_r Q)$. On this set the same two reservoir samples agree and the same third sample differs. □

Lemma 14.2 (Public fresh-reset pruning)**.** *Fix a public state and public blocks $B_1, \dots, B_R$ chosen before exposing fresh equality gates. Let $N = \sum_r |B_r|$. Suppose each gate satisfies*

$$\mathbb{P}[G_r \mid \textit{previous transcript}] \le \exp(-\eta |B_r|)$$

and that, after the public blocks are fixed and terminal value atoms are excluded, the auxiliary non-value transcript has at most $\exp(C_{\mathrm{aux}} N)$ possible atoms. If $\eta > C_{\mathrm{aux}}$, then the total mass of no-exit reset windows of incidence N is at most

$$\exp(-(\eta - C_{\mathrm{aux}})N).$$

Proof. For one fixed auxiliary transcript atom, the conditional gate probabilities multiply to at most

$$\prod_r \exp(-\eta |B_r|) = \exp(-\eta N).$$

There are at most $\exp(C_{\mathrm{aux}} N)$ such atoms. The union bound gives the claim. □

Theorem 14.3 (Fixed-scale old-incidence cap)**.** *Let $O_1, \dots, O_R \subseteq U$ be public old blocks of one active scale and one public sample-label signature ℓ for the ordered active triple/reservoir labels, and let $N = \sum_r |O_r|$. The window is run under the normal-form rule of Theorem 13.6: high-symbol common old values are not exposed by the old-window ledger. Before any nonterminal continuation from an old block, the proof reveals only the finite non-value data needed to choose the close-pair role and then either registers the block incidences or exits by high revisit. Let $M(i)$ be the cumulative number of low-revisit registered incidences of coordinate i on the current branch since i entered the old support, capped at the revisit threshold. These counters are not reset when the old support grows. The signature ℓ is fixed throughout the window; if a transition changes the active sample-label signature before the incidences are registered, the current bounded queue is charged to that transition and cleared instead of being inherited. Then one of the following occurs:*

(1) *a low-symbol old value is exposed and the branch immediately terminalizes as in Theorem 13.6;*

(2) *high revisit: adding the current block would make some $M(i) > Q$, and the current close-pair coordinate is routed by Lemma 13.3 into terminal value-shadow branches, a charged pruned complement, or a public low-collision hash coordinate;*

(3) *low-revisit continuation: every $M(i) \le Q$, no high-symbol old value is exposed, and the total non-value old transcript on the branch is bounded by*

$$C_{\mathrm{old}}(Q, r_*) \sum_{i \in U} M(i) \le C_{\mathrm{old}}(Q, r_*)\, Q|U|.$$

Thus high-symbol old returns have no alphabet-dependent value-entropy charge, and their remaining role, pattern, scale, and incidence data are paid by the old-incidence potential.

Proof. The signature ℓ is part of the public old-window state, so all equality patterns, role vectors, and common-symbol vectors below refer to the same ordered active labels. If a later transition replaces these labels before the window is charged, it is treated as the hard exit which clears the bounded queue; the old-pattern atom for ℓ is not reused for the new labels.

At a reached old block, first apply Theorem 13.6. If the common old-symbol vector has entropy at most $h|O_r|$, its value selector is terminal-local and the branch stops, giving alternative (1). Otherwise the branch is high-symbol and the value vector is not a child selector.

The only selectors exposed before continuation are the equality-pattern, role, scale, reservoir-label, and incidence data listed in the old rows of the global charge table. By definition of $C_{\mathrm{old}}(Q, r_*)$, these cost at most $C_{\mathrm{old}}(Q, r_*)$ per registered old incidence. If adding the incidences of O_r would make some counter exceed Q, the current close-pair occurrence of that coordinate satisfies the hypothesis of Lemma 13.3; it yields terminal coordinate value pins, a charged pruned complement, or a public low-collision hash coordinate, which is alternative (2). The current block is then charged to that exit and is not counted as a further low-revisit continuation.

Otherwise register O_r, increasing $M(i)$ by one for each $i \in O_r$. If no counter exceeds Q, then

$$\sum_r |O_r| = \sum_{i \in U} M(i) \le Q|U|.$$

The displayed cost bound follows. Since high-symbol values were never exposed, this low-revisit continuation contributes only old-incidence transcript, and the weighted debit in $\Phi_{\mathrm{old}} = B_o|U| - N_{\mathrm{old}}$ pays it. When these incidences also create a continuing old active child, the same debit first reserves its rank and scale balances. The remaining raw capacity condition is $N_{\mathrm{old}} \le Q\omega_{\mathrm{old}}|U| < B_o|U|$ in the constant hierarchy. $\square$

Short old windows are still concatenated so that a branch cannot restart the same fixed cost indefinitely without increasing the cumulative old-incidence counter.

Proposition 14.4 (Delayed old-window queue)**.** *Fix an old support U, a dyadic old scale, and a public sample-label signature ℓ for the ordered active triple/reservoir labels. The proof maintains one public queue $\mathcal{Q}(\ell, U, \mathrm{scale})$ of old blocks of that scale and signature. If the old support later grows from U to U^+ while ℓ is unchanged, the same queued blocks are relabelled as a queue for $(\ell, U^+, \mathrm{scale})$; no new coordinate is added to those blocks and no selector is re-exposed. The old-pattern atom is refined nestedly for the same signature: the atom on U^+ is required to restrict to the already paid atom on U, and only the pattern on $U^+ \setminus U$ for the same labels is newly exposed. If a transition changes the active sample-label signature before the queue is registered, the bounded unpaid queue is assigned to that transition's local charge and cleared; it is not inherited by the new signature. A bounded short window is appended to the queue and is not charged separately. The revisit count of a coordinate is cumulative over the whole queue, not over the last short window alone. Once the queued incidence*

$$N(\mathcal{Q}) = \sum_{O \in \mathcal{Q}} |O|$$

reaches the fixed old-window threshold, Theorem 14.3 is applied to the concatenated queue. The queue is cleared after its incidences are registered and debited in N_{old}, or earlier if a terminal, hash, scale, or high-revisit exit occurs. A fresh promotion which only enlarges the old support does not clear the queue if the signature is unchanged; it relabels the same queued blocks as above. If a fresh, hash, copy, scale, or birth transition is instead used as the hard exit paying the bounded unpaid queue or changing the signature, that bounded incidence is charged to the hard exit before the queue is cleared.

Consequently the delayed queue has uniformly bounded unpaid incidence between two charged exits for each signature, and all nonterminal delayed incidences are eventually debited in N_{old} or assigned to the signature-changing hard exit.

Proof. All blocks in the queue are public and have the same dyadic scale and the same public sample-label signature ℓ. Monotonic growth of U does not change the old coordinates already appearing in the queue. The nested-pattern convention from Proposition 3.7 makes the consistency statement literal: if P_U is the already paid five-state atom on U, then a later atom on U^+ is of the form

$$P_U \cap P_{U^+\setminus U}.$$

where both atoms refer to the same signature ℓ. Thus its restriction to every queued block contained in U is exactly the previously paid atom, and relabelling a queue from U to U^+ preserves the hypotheses of Theorem 14.3. The only new entropy is the five-state pattern on the newly promoted coordinates for the same labels, which is charged at the fresh promotion or old-incidence step which introduced them.

Concatenating short windows therefore preserves both publicness and scale. Coordinate multiplicities are counted in the concatenated list; hence high revisit is triggered by cumulative, not local, multiplicity. If the queue has not reached the fixed threshold, its unpaid incidence is bounded by that threshold. A pure growth of U with unchanged signature only changes the label of the same queue and does not create a new unpaid window. If the active signature changes, the old equality atom is no longer meaningful for the new labels, so the queue must be cleared first: if it has reached threshold, apply Theorem 14.3; otherwise charge its bounded unpaid incidence to the signature-changing hard exit. When the queue reaches the threshold, the old-window theorem either gives a terminal or high-revisit exit, or registers and debits the queued incidences in N_{old}. If some other hard exit is declared before threshold, the bounded unpaid incidence is assigned to that exit's local charge. After the queue is cleared, no old incidence from it is charged again. □

15 Bounded Active Reservoir

Definition 15.1 (Bounded active reservoir convention)**.** *Nonterminal states retain only a bounded active reservoir of samples. Value pins terminalize immediately as multi-sample shadows. A hash return may temporarily realize three conditioned copies, but after paying the public copy tax it retains only the active triple/pair/reservoir required for the next transition.*

This prevents exponential sample-label growth under repeated conditioned copying.

Lemma 15.2 (Reservoir marginalization is law preserving)**.** *Let a transition create auxiliary sample labels only to certify a public event or to realize conditioned copies. Let L be the retained active labels after the transition, and suppose that every future nonterminal selector and every nonterminal state variable is measurable with respect to the coordinates of L and the public transcript. Replacing the full conditional law on all temporary labels by its marginal law on L preserves the distribution of every future selector and of every projected terminal shadow involving retained labels. It also cannot increase the number of retained sample labels or the entropy of any retained public key.*

Proof. This is the elementary marginalization identity. For every event E measurable with respect to the retained labels and the public transcript, its probability under the full conditional law equals its probability under the marginal on those labels. Hence all later public decisions, which by hypothesis are such events or deterministic functions of such events, have the same law after the auxiliary labels are erased. Terminal constraints involving erased labels have already been projected or terminalized before erasure; therefore no future terminal shadow can require them. Finally, deterministic erasure and marginalization cannot create new retained labels and cannot increase Renyi or Shannon entropy of a retained transcript key. □

Proposition 15.3 (Reservoir labels stay bounded)**.** *Under Definition 15.1, the number of active sample labels in every nonterminal node is bounded by an absolute constant. Terminal shadows satisfy*

$$R(\Theta) \le T(\Theta) + O(1),$$

with the $O(1)$ term accounting only for the fixed active reservoir.

Proof. Induct over nonterminal transitions. Fresh, coordinate-cylinder, scale, and old transitions use the current bounded reservoir. A value pin stops and is counted as a multi-sample shadow, so it is not carried forward. A hash return may realize three conditioned copies, but after the public copy tax is paid the auxiliary labels are analysis-only data and are not retained. Lemma 15.2 shows that erasing those labels preserves the law of all later selectors. The next state keeps only the triple, pair, or bounded reservoir needed for the next transition.

Every nonempty terminal sample label pins at least one coordinate, giving $R(\Theta) \le T(\Theta)$ apart from the fixed bookkeeping labels already present in the reservoir. □

16 The Delayed-Window Potential

Define a finite-state potential

$$\Phi = \Phi_{\text{rank}} + \Phi_{\text{fresh}} + \Phi_{\text{old}} + \Phi_{\text{scale}}.$$

The accounting used below is global in the current unpinned ambient coordinate set $[k]$, not only in the current active block. Let m_{act} be the total rank of the active blocks currently carried by the state, after deleting coordinates already pinned in the public terminal projection; in single-block modes this is just the active block size. The term Φ_{rank} is the explicit form of the "initial active-rank" account used to birth the first active blocks and any DRC/split child whose size is a fixed fraction of its parent active rank. Let I_{old} be the raw cumulative old-window incidence count already registered on the current branch, capped by the high-revisit threshold through the coordinate counters $M(i) \le Q$. After the fixed-window parameters q, h, Q, r_* are chosen, let $C_{\text{fr}}(q)$, C_{hash}, and $C_{\text{old}}(Q, r_*)$ denote the finite transcript constants specified in Section 18, and put

$$\omega_{\text{old}} := C_{\text{old}}(Q, r_*) + B_{\text{rk}} + B_s + C_{\text{hash}} + 10.$$

The old potential uses the weighted debit $N_{\text{old}} := \omega_{\text{old}} I_{\text{old}}$, not the raw incidence count. Choose constants

$$B_o > Q\,\omega_{\text{old}} + C_{\text{aux}} + 10, \qquad B_f - B_o > C_{\text{fr}}(q) + C_{\text{hash}} + B_{\text{rk}} + B_s + 10.$$

Put

$$\Phi_{\text{rank}} = B_{\text{rk}} m_{\text{act}}, \qquad \Phi_{\text{fresh}} = B_f |[k] \setminus U|, \qquad \Phi_{\text{old}} = B_o|U| - N_{\text{old}}, \qquad \Phi_{\text{scale}} = B_s \sum_{\text{active } J} |J|.$$

The potential is used as an account balance, not as an unlabelled sum of all historical work. When a new active scale is born, its $B_s|J|$ scale budget is transferred from an existing recurrence account:

the initial active-rank balance, a parent active-rank drop, the parent active-scale balance, a fresh drop, or registered old incidences. Terminal shadows do not fund nonterminal scale births, and a DRC certificate pays selector entropy only; any scale deposit for the DRC-produced block is drawn from the parent active rank/scale balance or from the fresh/old progress which created the block. Fixed hash returns have size $h < m_0$ in the hierarchy, so they are bounded-scale returns unless their pending token is assigned to one of the existing hard-exit accounts before a larger child is entered. Thus a birth relabels already paid recurrence budget as rank/scale budget and is not allowed to create additional recurrence potential for free. The branch quantity $\Delta\Phi$ below is the charged expenditure from this account after such transfers, not the raw sum of every temporary increase and decrease in the displayed formula. When a DRC or split transition replaces an active rank m by a child block of size at most $\theta_{\mathrm{rk}} m$ with $\theta_{\mathrm{rk}} < 1$, the transfer into the child's scale balance, while keeping the child's inherited rank balance, is taken from the rank drop $B_{\mathrm{rk}}(m - \theta_{\mathrm{rk}} m)$ before any residual selector cost is charged. Thus consuming old incidences decreases Φ_{old} by the weighted debit, large enough to reserve child rank/scale balance and pay the local old transcript. Promoting any public fresh set $F \subseteq [k] \setminus U$, including residual coordinates outside the current active block, to old support changes the first two terms by

$$-B_f|F| + B_o|F| = -(B_f - B_o)|F|,$$

so promotion is a genuine potential drop. The high-revisit rule gives $I_{\mathrm{old}} \leq Q|U|$ on nonterminal low-revisit branches, hence $N_{\mathrm{old}} \leq Q\omega_{\mathrm{old}}|U| < B_o|U|$ before a terminal or bounded-support exit occurs.

Lemma 16.1 (One-step account inequalities)**.** *After the constants in Section 18 are fixed, the following deterministic inequalities hold for every nonterminal step. Let $\varepsilon_{\mathrm{hash}} \in \{0, 1\}$ indicate whether the step receives the unique hash token assigned by Proposition 9.7.*

(i) *If a fresh step promotes a nonempty set $F \subseteq [k] \setminus U$ and births an active child $J \subseteq F$, then*

$$(B_f - B_o)|F| \geq (B_{\mathrm{rk}} + B_s)|J| + C_{\mathrm{fr}}(q)|F| + \varepsilon_{\mathrm{hash}} C_{\mathrm{hash}} + |F|.$$

The same inequality without the child-balance term $(B_{\mathrm{rk}} + B_s)|J|$ applies when no child is born.

(ii) *If an old step registers $n \geq 1$ old-coordinate incidences and births an active child J with $|J| \leq n$, then*

$$\omega_{\mathrm{old}} n \geq (B_{\mathrm{rk}} + B_s)|J| + C_{\mathrm{old}}(Q, r_*)n + \varepsilon_{\mathrm{hash}} C_{\mathrm{hash}} + n.$$

(iii) *If a parent active rank m creates a public DRC/split child J with $|J| \leq \theta_{\mathrm{rk}} m$, then*

$$B_{\mathrm{rk}}(m - |J|) \geq (B_s + C_{\mathrm{rk}} + 1)|J| + \varepsilon_{\mathrm{hash}} C_{\mathrm{hash}}.$$

The child keeps the inherited $B_{\mathrm{rk}}|J|$ rank balance; only the displayed rank drop is spent.

(iv) *If a deletion-trapped scale step replaces an active block of size m by blocks of total size at most $4\sigma m$, then its scale drop is at least*

$$B_s(1 - 4\sigma)m.$$

For $m \geq m_0$ this drop dominates the fixed additive selector costs and any single assigned C_{hash} token by the choice of m_0.

Proof. For (i), use $|J| \le |F|$, $\varepsilon_{\text{hash}} \le 1$, and

$$B_f - B_o > C_{\text{fr}}(q) + C_{\text{hash}} + B_{\text{rk}} + B_s + 10.$$

For (ii), use $|J| \le n$, $\varepsilon_{\text{hash}} \le 1$, and

$$\omega_{\text{old}} = C_{\text{old}}(Q, r_*) + B_{\text{rk}} + B_s + C_{\text{hash}} + 10.$$

For (iii), $|J| \le \theta_{\text{rk}} m$ and the hierarchy gives

$$B_{\text{rk}}(1-\theta_{\text{rk}})m > (B_s + C_{\text{rk}} + C_{\text{hash}} + 10)\theta_{\text{rk}} m \ge (B_s + C_{\text{rk}} + 1)|J| + \varepsilon_{\text{hash}} C_{\text{hash}},$$

using $|J| \ge 1$ whenever a child is born. The last item is exactly Lemma 10.9 together with the lower bound imposed on m_0. □

Proposition 16.2 (Global charge summation)**.** *For the finite truncation, suppose every reached nonterminal selector charge c_v is decomposed as*

$$c_v = c_v^{\text{sh}} + c_v^{\text{fr}} + c_v^{\text{old}} + c_v^{\text{sc}} + c_v^{\text{hash}},$$

with the following ledger certificates on each retained branch:

(i) *terminal-shadow charges satisfy*

$$\sum_v c_v^{\text{sh}} \le C_{\text{sh}}(T(\Theta) + R(\Theta));$$

(ii) *fresh charges are attached to disjoint public sets newly promoted from $[k] \setminus U$ into U;*

(iii) *old charges are attached to old-coordinate incidences debited in N_{old}, and no coordinate is used after its revisit counter reaches the threshold Q unless the branch exits through the high-revisit rule;*

(iv) *scale charges at an active scale m occur only on a child whose next scale is at most θm for some fixed $\theta < 1$, and every new active block J receives a paid rank/scale certificate from the parent active-rank drop, the parent active-scale account, fresh progress, or old-incidence account which created it. In the fresh/old-source cases this certificate includes $(B_{\text{rk}} + B_s)|J|$. Terminal shadows and DRC selector certificates are not recurrence-potential sources;*

(v) *every nonterminal hash-window charge is either terminal, or is carried as one pending token of value at most C_{hash} and assigned to the first subsequent fresh promotion, old incidence, scale descent, or close-pair birth produced by the hash return; bounded hash-support exits are classified directly into fresh promotion or old-incidence registration by Proposition 9.5. This assignment is made before any child state or descendant hash window is entered from the returned residual block.*

Then, after increasing the constants in the potential hierarchy,

$$\sum_{v \text{ reached}} c_v \le C\,(T(\Theta) + R(\Theta) + \Delta\Phi)$$

on every retained branch, and hence the same inequality holds in expectation.

Proof. The terminal-shadow part is (i). The numerical inequalities used in the fresh, old, parent-rank, and scale paragraphs are exactly Lemma 16.1; the assertion that a hard exit pays at most one fixed hash token is Proposition 9.7, with Lemma 9.8 preventing an unpaid token from being passed to a descendant. For the fresh part, if disjoint public sets $F_a \subseteq [k] \setminus U$ are promoted to U, then

$$\Delta(\Phi_{\text{fresh}} + \Phi_{\text{old}}) = \sum_a (B_f - B_o)|F_a|$$

before old incidences are counted. The definition of $C_{\text{fr}}(q)$ includes the bounded fresh-pattern and promotion transcript per promoted coordinate. If a fresh promotion creates a continuing active child $J_a \subseteq F_a$, the account first transfers $(B_{\text{rk}} + B_s)|J_a|$ into that child's rank and scale balances. The remaining spendable fresh drop is at least

$$(B_f - B_o)|F_a| - (B_{\text{rk}} + B_s)|J_a|,$$

which, by $B_f - B_o > C_{\text{fr}}(q) + C_{\text{hash}} + B_{\text{rk}} + B_s + 10$, pays the finite fresh transcript and any single bounded hash charge assigned to that nonempty fresh exit. The transferred rank and scale balance remains in the child potential and is not counted again as spent expenditure.

For old data, each charged role, pattern, bounded-support use, or old-window auxiliary transcript contributes at most $C_{\text{old}}(Q, r_*)$ per old-coordinate incidence. If an old exit registers n raw incidences and creates a continuing active child J, then $|J| \le n$. The old account debits $\omega_{\text{old}} n$ from N_{old}, transfers $(B_{\text{rk}} + B_s)|J|$ into the child's rank and scale balance, and leaves at least

$$\omega_{\text{old}} n - (B_{\text{rk}} + B_s)|J| \ge C_{\text{old}}(Q, r_*) n + C_{\text{hash}} + 10$$

of spendable old expenditure whenever a hash token is assigned to that nonempty exit. This pays $\sum_v c_v^{\text{old}}$ and the assigned token, while the transferred rank and scale balance remains in the child potential. On every interval on which U is fixed, the debit ΔN_{old} is the raw drop of Φ_{old} before such transfers. When U increases, the increase $B_o|F|$ in Φ_{old} is already included in the net fresh-promotion drop $(B_f - B_o)|F|$ above. The cap $N_{\text{old}} \le Q\omega_{\text{old}}|U| < B_o|U|$ keeps Φ_{old} nonnegative on nonterminal low-revisit branches.

For scale data, a deletion-trapped transition from m to at most θm has drop at least $B_s(1-\theta)m$ in the active scale potential. Thus all charges $O(m)$ after a scale has been born are bounded by a geometric series times the deposited active scale budget. By (iv), births are not free: before a new active block J is entered, its inherited rank and scale balances are assigned to the ledger which created it. Proposition 17.2 verifies the source in each case: the initial block is paid by the root active-rank budget; DRC/split children of an existing active rank are paid by the corresponding rank drop or by that parent's deposited scale balance; fresh births by the fresh drop; old births by registered old incidences. A terminal shadow stops the branch and creates no nonterminal active child. A DRC atom and role refinement pay selector entropy, but not recurrence potential; the scale deposit for a DRC-produced block is taken from the parent active-rank drop, parent active-scale balance, or from the fresh/old progress which routed to it. Fixed hash returns have size below m_0, and any later larger active child must receive its deposit from the hard exit to which the hash token is assigned. After increasing the corresponding ledger constants, the birth budget is already included in the preceding fresh, old, parent-rank, or parent-scale paragraph; the bounded hash paragraph pays only the fixed hash token before it is assigned to one of those hard-exit accounts. Later scale transitions only spend that already deposited amount, so a positive increase of Φ_{scale} is never treated as a free negative drop.

Finally, (v) says a nonterminal hash window is not a separate reusable budget. It is carried as a single pending charge of the returned block until the first hard exit supplied by that block, or it is a bounded-support exit classified as fresh or old by Proposition 9.5. When the first hard exit occurs, the token value is included in that exit's local charge c_v and is removed from the returned block's state before any child state or descendant hash window is entered. Lemma 9.8 gives the exact state invariant: deletion-trapped scale children and close-pair birth children start with no inherited hash debt. This is a disjoint assignment of already incurred entropy charges, not an operation which deletes entropy. Since each fixed hash window has cost at most C_{hash} and the assigned hard exits are charged in the terminal-shadow, fresh, old, scale, or birth paragraphs, adding C_{hash} to the corresponding ledger constant pays the hash part. Summing the five paragraphs gives the branchwise inequality. $\square$

Proposition 16.3 (Terminal cylinder potential compatibility)**.** *Let a retained branch start from a parent public state* $\mathsf{S}_{\mathrm{parent}}$ *and terminate with projected shadow*

$$\theta = ((J_s, a_s))_{s\in S}.$$

After all pending hash tokens, delayed old queues, and active-scale birth deposits on that branch have been assigned, let $\Delta\Phi$ *be the charged potential expenditure of the branch. For each* $s \in S$*, form the cylinder child* $\mathsf{S}_{\theta,s}$ *whose underlying code is the full parent value cylinder* $\{x : x_{J_s} = a_s\}$ *with* J_s *deleted. Its ledger/support data are obtained from the terminal public state on the branch by deleting* J_s *from the current ambient coordinate set, active coordinates, old support, queues, and incidence counters; non-value terminal atoms are not additional code restrictions. Then*

$$\Delta\Phi + \sum_{s\in S} \Phi(\mathsf{S}_{\theta,s}) \le R(\theta)\Phi(\mathsf{S}_{\mathrm{parent}}).$$

Proof. The potential account used in Proposition 16.2 is a spent ledger. Hash tokens are assigned before a later hash window can open, delayed old queues are either registered or charged to the hard exit which clears them, and Proposition 17.2 assigns every scale birth to the ledger which created it before the born block is entered. After these assignments, the branch has a spent amount $\Delta\Phi$ and a remaining terminal public ledger Φ_{leaf} with

$$\Delta\Phi + \Phi_{\mathrm{leaf}} \le \Phi(\mathsf{S}_{\mathrm{parent}}).$$

The following table is a summary of the transition-by-transition check of this account invariant. The arithmetic entries are the inequalities of Lemma 16.1, with the hash-token multiplicity controlled by Proposition 9.7. "Balance" means the remaining public potential available to cylinder children.

transition source	account action	compatibility effect
public deterministic choice	no non-public selector and no potential transfer	$\Delta\Phi$ and the balance are unchanged
terminal value/shadow pin	charged to $T(\Theta) + R(\Theta)$, not to potential	pinned coordinates are deleted in cylinder children
product-KL or collision exit	terminal pins go to shadow; low-collision outputs enter a labelled hash token	no active scale or old queue is born without a later assigned token
hash bounded support	fresh part is promoted, or old incidences are registered	the window closes and cannot be charged again
hash near-sunflower return	C_{hash} is carried as one token and assigned to the first hard exit	the token is paid before any child state or descendant hash window is entered
fresh promotion	transfers any fresh-born $(B_{\mathrm{rk}} + B_s)\|J\|$ rank/scale balance, then spends the remaining drop from $\Phi_{\mathrm{fresh}} + \Phi_{\mathrm{old}}$	promoted coordinates are disjoint, and transferred rank/scale balance remains in the child
old residual/window	debits N_{old}, transfers any old-born rank/scale balance, then spends the remaining debit; or exits by high revisit	delayed queues are registered or assigned before terminal counting
scale birth	rank/scale balance is certified from parent rank drop, parent scale, fresh drop, or old incidences	terminal shadows and DRC selector certificates do not create recurrence balance
scale descent	spends the deposited scale balance geometrically	the child has smaller active scale and no inherited hash debt
DRC/split block	the public DRC atom and role refinement pay selector entropy	any rank/scale balance comes from parent rank/scale or fresh/old progress
reservoir/copy/pruning	copy tax or pruning mass is paid locally	auxiliary labels and pruned branches are not inherited by cylinder children
bounded leaf	absorbed by D_0 in the finite base case	no recursive cylinder child with unpaid balance is created

Formally, the account invariant is imposed at token-free checkpoints. A nonterminal hash return may create one bounded pending token, but this is a temporary one-step debt attached to the returned residual block, not a new recurrence balance. By Lemma 9.8, the token must be assigned to the exit which leaves that residual block before any child state or descendant hash window is entered. Let S_t be the accumulated spent potential after the first t checkpoint-to-checkpoint transitions of the branch, and let B_t be the public recurrence balance still carried by the checkpoint state, including inherited rank/scale balances but excluding selector entropy already charged to $T(\Theta) + R(\Theta)$ or to the local proof-tree ledger. Initially

$$S_0 = 0, \qquad B_0 = \Phi(\mathsf{S}_{\mathrm{parent}}).$$

For each nonterminal checkpoint transition, Lemmas 16.1 and 9.8 give

$$S_{t+1} - S_t + B_{t+1} \le B_t$$

with any unique possible hash token from Proposition 9.7 included in the hard-exit charge which restores the next checkpoint. During the intermediate residual state immediately after a hash return, the ordinary checkpoint inequality is not asserted; equivalently one may view the branch as carrying a bounded overdraft of size at most C_{hash}. That overdraft is cleared at the next checkpoint and cannot be inherited by a child. Public deterministic choices have equality; terminal value pins and projected shadows add no recurrence child; copying and pruning charges are local transcript or discarded-mass charges and do not increase B_{t+1}; bounded leaves stop the recurrence and are absorbed by D_0. Induction on token-free checkpoints yields the account invariant

$$\Delta\Phi + \Phi_{\mathrm{leaf}} \le \Phi(\mathsf{S}_{\mathrm{parent}}).$$

The child code itself is the full value cylinder, so the non-value terminal transcript does not affect the cylinder-size estimate in Theorem 5.2. For the inherited ledger, deleting pinned coordinates from the terminal public state cannot increase the rank, fresh, old-incidence, or scale components of Φ. The rank and scale terms decrease because the child ambient and active blocks are smaller. A deleted coordinate outside U is removed from the current ambient $[k] \setminus U$, so the fresh term decreases by B_f. The only nontrivial case is the old term. If a deleted old coordinate i has raw revisit counter $m_i \le Q$, deleting it changes

$$\Phi_{\text{old}} = B_o|U| - \omega_{\text{old}} I_{\text{old}}$$

by

$$-B_o + \omega_{\text{old}} m_i \le -B_o + Q\omega_{\text{old}} < 0.$$

Deleting queued incidences is no worse: queued incidences which have not yet been debited were either assigned to the hard exit before terminal counting, or else are absent from both the terminal child and I_{old}. Thus each inherited cylinder child satisfies

$$\Phi(\mathsf{S}_{\theta,s}) \le \Phi_{\text{leaf}}.$$

Therefore

$$\Delta\Phi + \sum_{s \in S} \Phi(\mathsf{S}_{\theta,s}) \le \Delta\Phi + R(\theta)\Phi_{\text{leaf}} \le R(\theta)\Phi(\mathsf{S}_{\text{parent}}),$$

because $\Delta\Phi + \Phi_{\text{leaf}} \le \Phi(\mathsf{S}_{\text{parent}})$ by the account invariant, $R(\theta) \ge 1$, and $\Phi_{\text{leaf}} \le \Phi(\mathsf{S}_{\text{parent}})$. ☐

Lemma 16.4 (Selector entropy primitives)**.** *All selector estimates used in Proposition 16.5 reduce to the following elementary bounds.*

(i) *Let p be a finite law and $0 < \theta \le 1$. If $H_\theta = \{a : p(a) \ge \theta\}$, then a selector which chooses an atom of H_θ or the complement has entropy at most*

$$\log(1 + \lfloor \theta^{-1} \rfloor).$$

On an atom branch the value has conditional mass at least θ and hence is a fixed-cost projected tuple pin.

(ii) *Let $0 < \rho, \tau \le 1$ and put*

$$H = \{a : p(a) \ge \rho\tau\}, \qquad L = H^c.$$

The atom branches in H are fixed-cost terminal pins. If $p(L) < \rho$, then discarding L has pruning loss $-\log(1 - p(L)) \le 2\rho$ for $\rho \le 1/2$. If $p(L) \ge \rho$, then

$$\operatorname{Coll}(p(\cdot \mid L)) = \sum_{a \in L} \left(\frac{p(a)}{p(L)}\right)^2 \le \frac{\max_{a \in L} p(a)}{p(L)} \le \tau.$$

(iii) *If a stopped prefix selector outputs a word u with length $\ell(u)$ and every retained output satisfies*

$$-\log P[U = u] \le \lambda \ell(u) + b,$$

then

$$\mathbf{H}(U) \le \lambda\, \mathbb{E}\ell(U) + b.$$

For tuple-coordinate prefixes, deterministic repeats have zero conditional entropy and inconsistent repeats have zero mass; after discarding those repeats the number of entropy-bearing values is at most the projected shadow coordinate count $T(\Theta)$.

(iv) *A retained public set event of conditional mass at least ρ adds at most $\log(1/\rho)$ to a labelled hash transcript. If appending that set event would exceed the remaining B_{win} budget, the set event is not appended and the branch terminalizes on the already paid value prefix.*

(v) *At fixed density parameters, a public DRC atom on a block of size s has cost $O(s)$, and a three-role refinement on that block has entropy at most $s \log 3$.*

(vi) *For one labelled hash-window key K, if $\mathbf{H}(K) \leq B_{\mathrm{win}}$ and G is the good-key event with conditional probability at least $1/2$, then*

$$\mathbf{H}_3(K \mid G) \leq \mathbf{H}(K \mid G) \leq 2B_{\mathrm{win}} + O(1).$$

Thus the three-copy Renyi tax is $2\mathbf{H}_3(K \mid G) + O(1) = O(B_{\mathrm{win}})$, and the exact split exposure on a fixed h-block costs at most

$$\log \sum_{r \leq 6\beta h} \binom{h}{r} \leq h \log 2.$$

(vii) *With Q, r_*, the old scale, and the sample-label signature fixed, the continuing old-window selectors have a finite per-incidence alphabet once high-symbol value vectors are not exposed. Hence their total entropy is at most $C_{\mathrm{old}}(Q, r_*)n$ on a branch registering n old incidences.*

Proof. For (i), $|H_\theta| \leq \theta^{-1}$. Item (ii) is the usual heavy-or-light decomposition; the pruning bound is $-\log(1-x) \leq 2x$ for $0 \leq x \leq 1/2$. Item (iii) follows by averaging the displayed pointwise prefix bound. The projection statement is Lemma 8.3. Item (iv) is the definition of the stopped per-labelled-window transcript. Item (v) is Lemma 10.1 and the bound $|\{XY|Z, XZ|Y, YZ|X\}^s| = 3^s$. For (vi), Renyi entropy decreases with order and conditioning on G of mass at least $1/2$ gives

$$\mathbf{H}(K \mid G) \leq \mathbf{H}(K)/\mathbb{P}(G) + \log 2 \leq 2B_{\mathrm{win}} + O(1);$$

Lemma 8.5 gives the factor $2\mathbf{H}_3$ for three copies. Item (vii) is the finite-alphabet accounting in Theorem 14.3: the continuing data consist only of sample-label signature, equality pattern, role, scale, bounded reservoir labels, and queue-flush markers, all charged per registered incidence before the revisit cap. □

Proposition 16.5 (Local selector entropy certificates)**.** *Run the finite truncation of the normal-form transition system. At a reached nonterminal node v, let E_v be the event that the branch reaches v and let Γ_v be the first non-public child selector after all deterministic public tie-breaks at v have been applied. Let ℓ_v^{pr} be the pruning-normalization loss at v, equal to $-\log q_v$ on a retained pruning step of retained conditional mass q_v and equal to 0 otherwise. There are nonnegative local charges*

$$c_v = c_v^{\mathrm{sh}} + c_v^{\mathrm{fr}} + c_v^{\mathrm{old}} + c_v^{\mathrm{sc}} + c_v^{\mathrm{hash}}$$

such that

$$\mathbf{H}(\Gamma_v \mid E_v) + \mathbb{E}[\ell_v^{\mathrm{pr}} \mid E_v] \leq \mathbb{E}[c_v \mid E_v]$$

for every reached node, and on every retained branch these charges satisfy the five ledger certificates of Proposition 16.2.

Proof. The elementary entropy estimates used in the rows are collected in Lemma 16.4. The following table gives the selector and the local entropy bound. All coordinate identities, block orders, and tie-breaks not shown in the second column are $\mathcal{F}_v$-measurable and therefore have zero conditional entropy.

transition	Γ_v and pruning loss	entropy/loss bound	ledger
public choice	deterministic block, scale, or tie-break	0	none
terminal atom/value pin	sample labels and pinned coordinate values	$C(T_v + R_v)$	shadow
low common-symbol entropy	$A_J = X_J = Y_J$ on the terminal block	$\mathbf{H}(A_J \mid E_v) \le h\lvert J\rvert$	shadow
collision exit	heavy atoms, light complement, or one hash coordinate	$O(1)$ plus possible pruning/hash transcript	sh./pr./hash
product-KL exit	interleaved prefix, tail/prune bit, or value atom from (A_j, B_j)	$\lambda\lvert\text{prefix}\rvert + O(1)$, or terminal value cost	shadow/hash
hash window	public sample label, transcript, split exposure, and copy key	$B_{\text{win}} + 2\mathbf{H}_3(K) + O(h)$	hash, then assigned
fresh residual	fresh equality pattern/support promotion	$C_{\text{fr}}(q)\lvert F\rvert$	fresh potential
old residual/window	sample-label signature, five-state pattern, role vector, incidence data; no high-symbol values	$\le \Delta N_{\text{old}}$ after the fixed old debit is applied	old potential
low-symbol old terminal	old common-value vector on O'	$\mathbf{H}(A_{O'} \mid E_v) \le h\lvert O'\rvert$	shadow
scale split/descent	DRC atom, exact split set, and role refinement on an active block	$O(\lvert J\rvert) + C_\sigma\lvert J\rvert + \lvert J\rvert \log 3$	rank/scale birth/drop
reservoir/copy labels	bounded active sample labels; auxiliary copies not retained in terminal data	$O(R_v)$ plus paid copy tax	reservoir/hash

We verify the rows in the order in which they can occur. Public rows have zero charge by the definition of the public sigma-algebra. Terminal atoms, terminal tuple pins, immediate value pins, and low common-symbol exits are terminal projected shadows. The cross/high-collision exits of Lemma 6.9 are heavy/light routings: their heavy branches are terminal projected shadows, their small complements are charged pruning branches, and their large complements are actual low-collision hash coordinates. The terminal value entropy is exactly the entropy of the pinned coordinates and the sample labels retained in the shadow. A one-coordinate heavy pin has fixed selector entropy by Lemma 16.4(i), while a terminalized prefix is paid by the amortized prefix estimate Lemma 16.4(iii). Hence terminal value data is charged to c_v^{sh} and contributes to $T(\Theta) + R(\Theta)$.

The product-KL row is Proposition 12.6. The prefix value is either amortized, with cost at most $\lambda\lvert\text{prefix}\rvert$, or the branch is routed through the first-crossing heavy/light scan using the previous amortized prefix. If no positive-KL or first-crossing stop occurs, the full tuple prefix is amortized and terminal. Positive-tail and first-crossing light-complement events have fixed cost when retained in the hash transcript; if they would overflow the window they are not appended. Any low-collision output is the actual conditional law of one tuple coordinate under the paid transcript. Thus the row is charged to terminal shadow, pruning, or to the hash ledger, never to an unpaid pair-alphabet selector. At a retained pruning step the child selector itself is deterministic after the public set is known, but the normalization loss $\ell_v^{\text{pr}} = -\log q_v$ is nonzero; the estimates in Lemma 16.4(ii)–(iv), equivalently the corresponding estimates in Lemmas 6.8 and 6.7, put that loss into the same local charge.

The hash row is Theorem 9.2 together with Propositions 9.5 and 9.6. The sample label is public, and the selected coordinate block is a deterministic function of the paid window transcript. The local charge consists of the window transcript entropy, the order-three Renyi copying tax for that label's conditional law, and the exact split exposure, bounded by Lemma 16.4(vi). If the block is returned to residual mode, this charge is carried as one pending token and assigned to the first subsequent terminal, fresh, old, scale, or birth exit before another hash window can be opened from that returned block or from a child born inside it; a bounded leaf or terminal shadow pays it directly.

Fresh residual charges are supplied by Lemma 10.3. The promoted fresh coordinates form a public set F disjoint from all earlier fresh promotions, and the finite pattern/role transcript costs at most $C_{\mathrm{fr}}(q)|F|$. This is c_v^{fr}.

Old residual and old-window charges are supplied by Lemma 13.5, Proposition 13.7, Theorem 13.6, Theorem 14.3, and Proposition 14.4. Low-symbol old values terminalize and are charged to the shadow row. High-symbol old values are not selectors of the continuing old-window tree. The remaining sample-label signature, equality-pattern, role, scale, reservoir-label, and incidence data are bounded by $C_{\mathrm{old}}(Q, r_*)$ per registered old incidence. The weighted debit ω_{old} also reserves any old-born rank and scale balances and any single hash token assigned to the nonempty old exit. The delayed queue prevents the same bounded short window for a fixed signature from being repeatedly recharged without increasing the raw counter and hence N_{old}; if the signature changes, the bounded queue is charged to that hard exit before it is cleared. When a high-revisit branch is routed by Lemma 13.3 to a one-coordinate hash output rather than to terminal value pins or a charged pruned complement, the selector has fixed cost $O(1)$ by Lemma 16.4(ii). This cost is charged to the same old-incidence/high-revisit account which detects the cap; if the coordinate later participates in a hash return, the window transcript and copy cost are separately paid by the single hash token assigned to the next hard exit. Thus queue increments before a completed window are not hidden inside the final token, and the complement of a heavy old value is never silently discarded.

Scale rows come from the DRC and split-exposure steps in residual repair and from deletion-trapped descent. Lemma 10.1 gives a DRC atom of cost $O(|J|)$ because the density parameters are fixed. The exact sparse split exposure costs at most $C_\sigma|J|$, and role refinement costs $|J|\log 3$. Lemma 10.9 gives the geometric scale drop, while Proposition 17.2 supplies the birth certificate for every new active block. When the block is born by replacing an active rank-m state by a fixed-fraction DRC or split child, the certificate is drawn from the active-rank drop before the child rank potential is inherited.

Finally, reservoir and conditioned-copy labels are controlled by Proposition 15.3 and Lemmas 8.4 and 8.5. Auxiliary copy labels are paid by their copy tax and are not retained in the terminal transcript. The public fresh-reset pruning loss of Lemma 14.2 is included in the auxiliary constant C_{aux} and hence in the hard-exit charge which clears that window. This proves the displayed conditional entropy inequalities and also verifies the five branchwise ledger certificates required by Proposition 16.2. $\square$

Theorem 16.6 (Delayed-window shadow-potential assembly)**.** *Run the finite truncation of the transition system from Definition 3.5 and Proposition 3.8, enforce Definition 11.1, use Theorem 14.3, use Definition 15.1, use the product-KL exit of Proposition 12.6, and choose the constants as in Section 18. Assign every new active scale the birth certificate specified in Definition 17.1; for the finite-state system this assignment is verified in Proposition 17.2. The branch invariants are those of Proposition 3.7. Then the complete proof tree has pruning loss*

$$L_{\mathrm{pr}} = \sum_{\text{pruning steps on the branch}} -\log(\text{retained mass}),$$

with pruning steps interpreted as in Proposition 3.10, and satisfies

$$\mathbf{H}(\Theta) + \mathbb{E}L_{\mathrm{pr}} \le C\,\mathbb{E}[T(\Theta) + R(\Theta)] + C\,\mathbb{E}[\Delta\Phi].$$

Consequently Theorem 5.2 and Theorem 4.2 give a constant-base recurrence.

Proof. Let Γ be the full terminal leaf transcript of the finite truncation: it records every non-public child selector, every exact-exposure atom, every role-refinement atom, and the terminal projected shadow before unpaid analysis-only data are projected away. Proposition 3.9 gives the retained transcript entropy, and Proposition 3.10 adds exactly the accumulated normalization loss of charged pruning steps. Thus

$$\mathbf{H}(\Theta) + \mathbb{E}L_{\mathrm{pr}} \leq \sum_{v} \mathbb{P}(E_v) \big(\mathbf{H}(\Gamma_v \mid E_v) + \mathbb{E}[\ell_v^{\mathrm{pr}} \mid E_v]\big).$$

By Proposition 16.5, the local charges satisfy

$$\mathbf{H}(\Gamma_v \mid E_v) + \mathbb{E}[\ell_v^{\mathrm{pr}} \mid E_v] \leq \mathbb{E}[c_v \mid E_v],$$

and their branchwise decomposition satisfies the five hypotheses of Proposition 16.2. Therefore Proposition 16.2, followed by expectation, gives

$$\mathbf{H}(\Theta) \leq C\,\mathbb{E}[T(\Theta) + R(\Theta)] + C\,\mathbb{E}[\Delta\Phi]$$

for a constant C depending only on the fixed thresholds. This estimate is obtained before any use of shadow descent.

It remains to check that the potential in Theorem 5.2 is the recurrence potential of the cylinder children, not the potential of an intermediate transition child. This is exactly Proposition 16.3: the cylinder children are formed from the terminal public state by deleting the pinned shadow coordinates, and their inherited potentials satisfy the recurrence-compatibility hypothesis in Theorem 5.2.

Now Theorem 5.2 converts the entropy estimate to a pointwise shadow atom and combines it with the stateful version of Theorem 4.2. The $T(\Theta), R(\Theta)$ terms are absorbed after D is chosen in Section 18. ☐

17 Active Scale Birth

It remains to prevent free increases of Φ_{scale}.

Definition 17.1 (Rank/scale birth certificate)**.** *Fix the rank and scale coefficients B_{rk}, B_s. Every new active block J is assigned a public certificate for the rank and scale balances it carries. If J is a fixed-fraction child of a parent active rank m, the certificate keeps the inherited $B_{\mathrm{rk}}|J|$ rank balance in the child and pays the selector and $B_s|J|$ scale balance from the parent rank drop or parent active-scale account. If J is born from a fresh or old hard exit, the certificate transfers*

$$(B_{\mathrm{rk}} + B_s)|J|$$

from the corresponding fresh/rank or old-incidence account into the child. The only allowed public sources are: initial active-rank, parent active-rank drop, parent active-scale, fresh/rank progress, or old-incidence. Terminal shadows are terminal and do not create nonterminal active children. Public DRC certificates pay selector entropy but are not independent sources of recurrence potential. Fixed hash-window returns are below the bounded-leaf scale m_0 unless their pending token is assigned to one of the listed hard-exit accounts before any larger active child is entered; by Lemma 9.8 the child itself is token-free.

Proposition 17.2 (Scale birth audit)**.** *Under the public DRC convention, immediate terminalization, public hash returns, and fixed-scale old-window reduction, Definition 17.1 holds for all active blocks produced by the finite-state system.*

Proof. Initial blocks are paid by the active-rank budget of the root state. A terminal shadow stops the branch, so it never pays for a continuing active scale. Residual DRC blocks and close-pair activation blocks produced from Lemma 10.1 have public selector cost $O(|J|)$ because all density parameters are fixed constants, but that selector cost is not a scale deposit. If such a block is a fixed-fraction child of a parent active rank m, the constant hierarchy makes the rank drop $B_{\mathrm{rk}}(m-|J|)$ large enough to reserve $B_s|J|$ and pay the selector cost while the $B_{\mathrm{rk}}|J|$ rank balance remains inherited in the child. If the block is instead produced by a fresh or old escape, the full $(B_{\mathrm{rk}}+B_s)|J|$ rank/scale balance is drawn from the corresponding fresh drop or old-incidence account. Near-sunflower activation split exposures are paid by the same active-rank certificate. Fresh residual blocks are paid by fresh/rank progress; in the bounded $|F| \le q$ case, Lemma 10.3 first promotes the nonempty split part F_{sp} before the fresh close-pair child is entered. Old residual blocks are paid by weighted old incidence, including the old-supported residual witness routing of Lemma 13.5. Pooled close-pair blocks are paid by the incidence window which produced them. High-revisit bounded-support blocks are paid by the repeated old incidences that triggered the high-revisit exit. One-coordinate KL increments are not active scale chains. A fixed hash return has size $h < m_0$ and is therefore bounded at the recurrence interface unless a later hard exit creates a larger active child; that later child is funded by the hard-exit account to which the hash token has been assigned, and enters token-free by Lemma 9.8. If a fixed hash block is ever treated as a child of a parent active rank $m \ge m_0$, the choice $m_0 > h/\theta_{\mathrm{rk}}$ gives $h \le \theta_{\mathrm{rk}} m$, so the parent-rank inequality of Lemma 16.1 applies as a separate fallback. This exhausts the block sources in the finite-state normal form without using terminal shadows or DRC selector certificates as independent recurrence-potential sources. □

18 Constant Hierarchy

The constants are chosen in the following order.

(1) Choose a cross-collision threshold $0 < \gamma_0 < 1$, then choose the activation collision threshold α and the residual density/hash survival threshold $0 < \beta < 1/48$ small enough for the σ below to satisfy $4\sigma < 1$. Choose a product-transfer tolerance $\epsilon > 0$ with $1/2 - 3\sqrt{\epsilon/2} > 0$. Put

$$\sigma := \max\left\{6\alpha,\ \frac{6\alpha}{(1-\gamma_0)^2},\ 6\beta\right\}$$

for Lemma 10.9. Fix the scale-budget coefficient B_s in Definition 17.1 large enough to pay the exact split exposure cost $\log \sum_{r \le \sigma m} \binom{m}{r}$ and the associated role-refinement cost on each active block of size m. Then choose $\tau \le \min(\alpha, \gamma_0^2, \beta)$ for the activation, collision-exit, and hash survival scans. Define

$$\theta_{\mathrm{rk}} := \max\{\sigma,\ (1-\tau)\alpha/4,\ \beta/4,\ 4\sigma\} < 1.$$

Positive-KL value pins have the fixed mass threshold $\rho\tau$, where $\rho = 1/2 - 3\sqrt{\epsilon/2}$. There is no constant-hierarchy condition involving the binary entropy of subsets of L: residual repair first promotes all possible fresh witnesses globally, and the remaining old-supported witnesses are paid by the old-incidence ledger rather than by exposing W_L.

(2) Choose the KL prefix threshold, an absolute fixed hash-window length h, and a cumulative hash transcript budget B_{win} so that Theorem 12.5 feeds only bounded-prefix coordinates into the fixed hash window while the cumulative budget remains available; amortized prefixes exceeding the remaining transcript budget terminalize as projected shadows. Definition 9.1 terminalizes any later atom of mass $\ge \beta$.

(3) Choose the residual support threshold q, the old-window queue threshold, the old revisit threshold Q, and reservoir size r_*. These determine the finite constants

$$C_{\mathrm{fr}}(q), \qquad C_{\mathrm{hash}}, \qquad C_{\mathrm{old}}(Q, r_*),$$

as follows. $C_{\mathrm{fr}}(q)$ is chosen as a per-promoted-coordinate constant large enough to pay the maximum fresh selector/transcript entropy over the finite list of residual transitions exposing at most q fresh coordinates. In particular, when $|F| \le q$ but only a nonempty split part F_{sp} is promoted, the whole $q \log 5$ five-state pattern exposure and the bounded close-pair routing produced from it are bounded by $C_{\mathrm{fr}}(q)|F_{\mathrm{sp}}|$. It excludes rank/scale deposits and hash tokens, which are paid separately by the displayed inequalities for $B_f - B_o$. The hash constant is taken explicitly as

$$C_{\mathrm{hash}} = C_0\big(B_{\mathrm{win}} + h + 1\big),$$

where C_0 is the absolute constant covering the proof-tree transcript entropy B_{win}, the Renyi copy tax $2\mathbf{H}_3(T_G \mid P_0) \le 4B_{\mathrm{win}} + O(1)$, the exact split exposure $\log \sum_{r \le 6\beta h} \binom{h}{r}$ (or the cruder $h \log 2$), and the bounded survival-scan bookkeeping in Theorem 9.2 and Proposition 9.5. $C_{\mathrm{old}}(Q, r_*)$ is the maximum old-window auxiliary transcript cost per old incidence before the revisit cap Q, including the sample-label signature, five-state pattern, three-role refinement, bounded reservoir labels, and the bounded queue-flush charge when the active signature changes. It excludes old-born rank/scale deposits and a possible assigned hash token, which are paid by the later choice of ω_{old}. All three are finite because q, h, Q, r_* and the window length are fixed constants.

(4) Choose the bounded-leaf scale $m_0 > \max(q, h/\theta_{\mathrm{rk}}, 4/\beta, 8/((1-\tau)\alpha))$ large enough that, for every scale $m \ge m_0$, the geometric scale drop $B_s(1 - 4\sigma)m$ also absorbs any fixed additive token assigned to that scale transition, including C_{hash} and the finite one-step scale selector constants. Then choose a crude constant D_0 large enough to dominate all states of active rank and block size below m_0.

(5) Choose the public fresh-reset gate parameter η larger than the finite auxiliary transcript constant C_{aux} in Lemma 14.2.

(6) Let C_{rk} be the finite per-coordinate selector constant for root activation, DRC row-block selection, sparse split exposure, and role refinement at the fixed thresholds above. Choose the active-rank coefficient B_{rk} so that

$$B_{\mathrm{rk}}(1 - \theta_{\mathrm{rk}}) > (B_s + C_{\mathrm{rk}} + C_{\mathrm{hash}} + 10)\theta_{\mathrm{rk}}.$$

Then choose the old-incidence capacity coefficient

$$\omega_{\mathrm{old}} := C_{\mathrm{old}}(Q, r_*) + B_{\mathrm{rk}} + B_s + C_{\mathrm{hash}} + 10, \qquad B_o > Q\,\omega_{\mathrm{old}} + C_{\mathrm{aux}} + 10,$$

and then choose the fresh coefficient B_f so that

$$B_f - B_o > C_{\mathrm{fr}}(q) + C_{\mathrm{hash}} + B_{\mathrm{rk}} + B_s + 10.$$

With this choice, every public residual support promotion in Lemma 10.3 has positive net potential drop and bounded hash-support promotion has a fresh certificate after reserving the rank and scale balances of the active child born from that promotion.

(7) Let C_* be the maximum of the finitely many row constants in the local certificate table of Proposition 16.5, after the choices above have fixed $q, h, Q, r_*, B_s, B_{\mathrm{rk}}, B_o, B_f$ and the window thresholds. Finally choose $A \geq C_*$ and

$$D \geq \max\{D_0, \exp(2C_*)\}$$

large enough to absorb the terminal shadow, reservoir, birth-certificate, and potential-drop constants in Theorems 4.2 and 5.2.

Proposition 18.1 (Feasibility of the constant hierarchy)**.** *The constants in Section 18 can be chosen so that all displayed inequalities and all fixed-window requirements used earlier in the proof hold simultaneously.*

Proof. The displayed requirements reduce to the following finite list.

First choose any fixed $0 < \gamma_0 < 1$, then choose $\alpha > 0$ so small that $24\alpha/(1-\gamma_0)^2 < 1/2$, and choose $\epsilon > 0$ with

$$\rho := 1/2 - 3\sqrt{\epsilon/2} > 0.$$

Next choose $0 < \beta < 1/48$ small enough that $24\beta < 1/2$, and put

$$\sigma := \max\left\{6\alpha, \frac{6\alpha}{(1-\gamma_0)^2}, 6\beta\right\}.$$

Then $4\sigma < 1$. Since $\sigma < 1/4$,

$$C_\sigma = \sup_{m \geq 1} m^{-1} \log \sum_{r \leq \sigma m} \binom{m}{r}$$

is finite. The scale-budget coefficient B_s is chosen to exceed the fixed multiple of $C_\sigma + \log 3$ used for split-set exposure and role refinement.

Choose

$$0 < \tau \leq \min(\alpha, \gamma_0^2, \beta).$$

Then

$$\theta_{\mathrm{rk}} := \max\{\sigma,\ (1-\tau)\alpha/4,\ \beta/4,\ 4\sigma\} < 1.$$

This single inequality is the value-pin condition used by activation, collision exits, and hash survival; the positive-KL value-pin threshold is the smaller fixed constant $\rho\tau$. It also removes the former binary entropy constraint on subsets of L, because residual repair no longer exposes a random old-supported set $W_L \subseteq L$ as an unpaid sparse selector.

Choose the KL prefix threshold and the cumulative transcript budget B_{win} for one fixed labelled hash window, with B_{win} larger than the fixed tail/pruning increments $\log(1/\rho) + O(1)$ and the corresponding positive-tail and first-crossing light-complement pruning charges. This is a per-window, per-sample-label budget. Multiple sample labels may open distinct windows on one branch; their transcripts are not concatenated into one global B_{win} account, and each completed or bounded window is paid separately by the hash-token matching ledger. The construction of Theorem 12.5 sends a coordinate into its labelled hash mode only when its prefix and tail/pruning transcript fits in the remaining budget of the current window. If a set-valued tail/pruning increment would overflow the window, it is not appended; the branch terminalizes on the already paid value prefix, and any larger prefix is terminalized as a projected shadow. If the budget closes before $2h$ distinct live coordinates are accumulated, the branch either has already terminalized on the projected prefix or is a bounded-support hash exit. Hence this step imposes no upper bound on B_{win} from later constants.

After q, h, Q, r_* and the old-window queue threshold are fixed, the constants $C_{\mathrm{fr}}(q)$, C_{hash}, and $C_{\mathrm{old}}(Q, r_*)$ defined above are finite. Since $4\sigma < 1$, choose $m_0 > \max(q, h/\theta_{\mathrm{rk}}, 4/\beta, 8/((1-\tau)\alpha))$ so large that $B_s(1-4\sigma)m_0$ dominates C_{hash} and the finite additive constants of a single scale transition, and then choose D_0 for the finitely many bounded active ranks and block sizes below m_0. The fresh-reset parameter is chosen so that

$$\eta > C_{\mathrm{aux}}.$$

Let C_{rk} be the resulting finite per-coordinate constant for active-rank DRC/split selectors. Choose

$$B_{\mathrm{rk}}(1-\theta_{\mathrm{rk}}) > (B_s + C_{\mathrm{rk}} + C_{\mathrm{hash}} + 10)\theta_{\mathrm{rk}}.$$

Then choose

$$\omega_{\mathrm{old}} := C_{\mathrm{old}}(Q, r_*) + B_{\mathrm{rk}} + B_s + C_{\mathrm{hash}} + 10, \qquad B_o > Q\,\omega_{\mathrm{old}} + C_{\mathrm{aux}} + 10$$

and

$$B_f > B_o + C_{\mathrm{fr}}(q) + C_{\mathrm{hash}} + B_{\mathrm{rk}} + B_s + 10.$$

These two inequalities are exactly the potential requirements used in Proposition 16.2: $B_o > Q\omega_{\mathrm{old}}$ keeps $\Phi_{\mathrm{old}} = B_o|U| - N_{\mathrm{old}}$ nonnegative before high revisit because $N_{\mathrm{old}} = \omega_{\mathrm{old}} I_{\mathrm{old}}$ and $I_{\mathrm{old}} \leq Q|U|$. The weighted debit ω_{old} pays old-incidence transcript, reserves any old-born rank and scale balances, and pays any single hash token assigned to that nonempty old exit, while $B_f - B_o > C_{\mathrm{fr}}(q) + C_{\mathrm{hash}} + B_{\mathrm{rk}} + B_s + 10$ pays every fresh promotion together with any bounded hash charge after reserving the fresh-born rank and scale balances assigned to it.

At this stage every local conditional-entropy estimate in Proposition 16.5 has a fixed finite coefficient. More explicitly, after the threshold choices above one may take

$$\begin{aligned} C_* = C_{\mathrm{abs}} \max\{&C_{\mathrm{sh}},\ C_{\mathrm{aux}},\ C_\sigma + \log 3,\ B_s,\ C_{\mathrm{rk}},\ C_{\mathrm{fr}}(q),\ C_{\mathrm{hash}},\ C_{\mathrm{old}}(Q, r_*),\\ &B_{\mathrm{rk}},\ \omega_{\mathrm{old}},\ B_o,\ B_f,\ r_*,\ 1\}, \end{aligned}$$

where C_{abs} is an absolute numerical constant and C_{sh} denotes the terminal projected-shadow constant from Theorem 4.2. This displayed maximum is not intended to optimize constants; it records the dependency graph. In particular,

$$\begin{aligned} C_{\mathrm{hash}} &= C_0(B_{\mathrm{win}} + h + 1),\\ B_{\mathrm{rk}} &= O((B_s + C_{\mathrm{rk}} + C_{\mathrm{hash}} + 1)/(1-\theta_{\mathrm{rk}})),\\ B_o &= O(Q(C_{\mathrm{old}} + B_{\mathrm{rk}} + B_s + C_{\mathrm{hash}} + 1) + C_{\mathrm{aux}} + 1),\\ B_f &= B_o + O(C_{\mathrm{fr}} + C_{\mathrm{hash}} + B_{\mathrm{rk}} + B_s + 1). \end{aligned}$$

Thus C_* may be very large, but it depends only on the fixed hierarchy and not on k, the alphabet sizes, or $|\mathcal{C}|$. It also does not depend on A or D. Choose

$$A \geq C_*, \qquad D \geq \max\{D_0, \exp(2C_*)\}.$$

Then the base cases below m_0 are dominated by D^j, and the shadow potential theorem may be used with $\lambda = c = C_*$, since

$$\log D \geq 2C_* = \lambda + c.$$

Every right-hand side above is already fixed at the moment it is used, and no later displayed requirement asks for an upper bound on B_o, B_f, A, D or a lower bound on α, ϵ, τ. $\square$

19 Interface Verification

The public-normal-form reduction above uses explicit interfaces for root activation, public DRC, residual support promotion, old-supported residual routing, hash windows, projected shadows, active reservoirs, global charge summation, and scale birth. Proposition 12.6 routes the product-KL excess branch through a genuine sample coordinate, so the verification never treats a low-collision law on the pair alphabet as a code coordinate.

Corollary 7.7 is used only as a bounded-alphabet and variable-profile check; it is not the alphabet-free mechanism. Its bound contains Q because slice rank pays for exposed alphabet values. In the normal-form proof, high-alphabet old values are not exposed as selectors: low-symbol value vectors terminalize locally, high-symbol old data is routed through incidence and scale ledgers, and product-KL creates only a one-coordinate pin or hash coordinate under the actual tuple law. Thus the alphabet size can enter only through terminal projected shadows or bounded local transcripts already counted in $T(\Theta), R(\Theta)$, not as a factor in the active-rank recurrence.

The fixed-window mechanism is therefore not a random restriction over the alphabet. A coordinate contributes to a hash key only if its whole value-prefix and tail/pruning transcript fits inside the fixed budget B_{win}. If the KL search reaches a long prefix, the framework stops on that value prefix and pays it as a projected shadow; it is not copied through the Renyi lemma. Thus the copy key has entropy at most B_{win} by construction, while the cost of a long alphabet-dependent value description is routed to $T(\Theta)$ instead of to the active-rank recurrence. This is the formal point at which the proposed framework departs from ALWZ-style random restriction accounting.

The old-supported sparse residual selector is no longer exposed as an unpaid atom of $W_L \subseteq L$. Lemma 10.3 first promotes all possible fresh witnesses globally, and Lemma 13.5 handles the remaining old-supported positive support by exposing only old-coordinate pattern data through the old-incidence ledger.

The global projected-shadow mass interface is supplied by Proposition 3.9, Proposition 16.5, and Theorem 5.2. The probability of reaching a local terminal choice is not discarded: it is part of the terminal leaf transcript before projection, and after merging leaves with the same projected shadow the mass of that shadow can only increase.

The product-KL exit creates only a one-coordinate terminal pin or hash coordinate. Later uses of the close-pair theorem enter either terminal shadow mode or the fixed-window hash ledger, and do not require a density increment from product KL.

The hash ledger is label-specific: a low-collision output from a tuple node is entered into the window of the actual public sample label which produced it. The fixed-window theorem then works with the conditional law of that one sample label and realizes the required independent copies by the Renyi copy lemma. Thus no step converts a low-collision statement for a mixed tuple alphabet, or for the reference law Q_E, into an unproved statement about the ambient code law.

20 Conclusion

Combining Theorem 16.6, Proposition 17.2, and the multi-shadow recurrence gives the conditional constant-base bookkeeping theorem of Theorem 1.1. The local exits are organized so that the only permitted outputs are public blocks, terminal projected shadows, potential drops, or pruning charges; private terminal transcripts are not allowed to enter the final counting. If the interface verifications recorded above are accepted without additional hypotheses, the front reduction in Section 2 transfers the resulting encoded code bound back to ordinary rank-k set families. Proposition 18.1 records the corresponding non-circular constant hierarchy.

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